Precalculus II

An Original Ten-Week Course Reader

Dr. Lee MATH& 142

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1 Angles, Radians, and the Unit Circle

A turning wheel connects angle, distance, speed, and position. This chapter develops those connections from a coordinate sketch to a model of a rider's height over time. Work through each example with pencil and paper. Pause at each check before reading its explanation at the end of the chapter.

Learning objectives. Describe signed rotations; convert between degrees and radians; distinguish arc length, displacement, angular velocity, and tangential speed; derive exact unit-circle coordinates; and connect circular motion with periodic graphs and a height model.

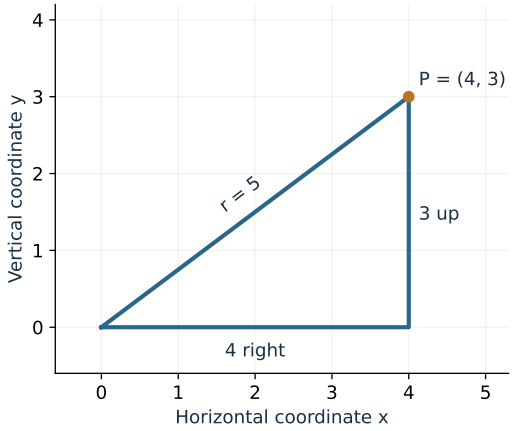
Online reading and interactive activities

Further reading: OpenStax, *Precalculus 2e*, 5.1 Angles and 5.2 Unit Circle: Sine and Cosine Functions.

1.1 Read a point before describing a turn

Imagine a marker on the rim of a turning wheel. We want to describe where the marker is, how far it has traveled, and how quickly it moves. These are different questions. Trigonometry connects them by treating an angle as a record of turning.

Begin with a coordinate plane. The origin is the point where the axes meet. In an ordered pair (x, y), the first number tells us horizontal position and the second tells us vertical position. Positive x means right; negative x means left. Positive y means above the horizontal axis; negative y means below it.



The coordinates describe two perpendicular moves. The distance from the origin is the hypotenuse.

A point and its distance from the origin

For $P = (4, 3)$, the Pythagorean theorem gives the distance r from the origin:

$$r^2 = 4^2 + 3^2 = 25, \qquad r = 5.$$

If we shrink this triangle until its hypotenuse is 1, every length is divided by 5. The new point is $(4/5, 3/5)$. It points in the same direction as $(4, 3)$, but lies on a circle of radius 1. This process is called *normalizing* the coordinates.

We will use fractions and square roots to describe some positions exactly. An expression such as $\sqrt{2}/2$ is an exact number. Its decimal approximation, about 0.707, is useful for estimating, but the rounded decimal loses some information. Both forms will appear in this module.

As you read, keep a sketch beside your calculations. A positive vertical coordinate should put the point above the horizontal axis; a distance should never be negative. These simple checks can catch many errors.

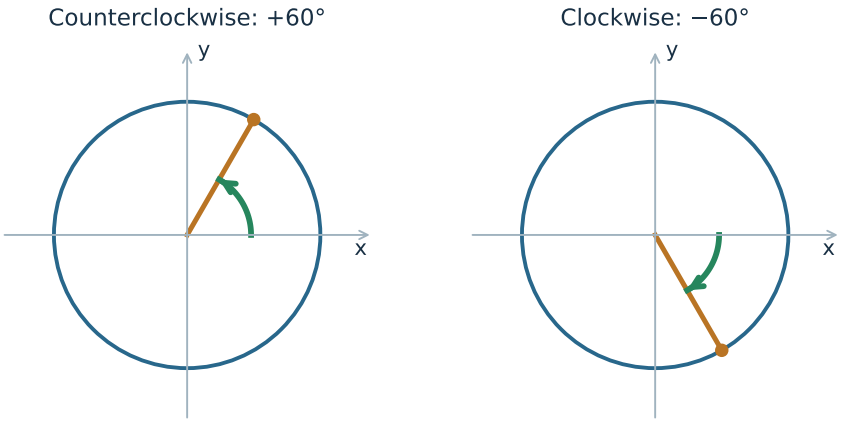
Check 1

A point is at $(-3, 4)$. Find its distance from the origin and divide both coordinates by that distance. Which signs should remain?

1.2 An angle records a signed rotation

An angle describes how a ray turns around its endpoint. A ray is a line that starts at one point and extends in one direction. In *standard position*, the endpoint is the origin and the starting ray points right along the positive x-axis. The final ray is called the *terminal ray*.

We use a sign to record the direction of turning. Counterclockwise turns are positive; clockwise turns are negative. A complete turn is 360°, a half-turn is 180°, and a quarter-turn is 90°. The Greek letter θ, pronounced "theta," often stands for an angle.



Begin at the positive horizontal axis. The sign tells you which way to turn.

Read the whole rotation

For 420°, turn counterclockwise through one full revolution and then another 60°:

$$420^\circ = 360^\circ + 60^\circ.$$

For -120°, turn clockwise through 120°. The terminal ray points down and left. You could reach that same ray by turning counterclockwise through 240°, but the two descriptions record different journeys.

A diagram usually shows the final direction more easily than it shows several complete turns. Do not

assume that an angle must be between zero and one revolution. A wheel may turn many times, and a negative angle is a perfectly valid description of motion.

Check 2

Where do $-90°$ and $450°$ end? Describe the direction and amount of turning for each.

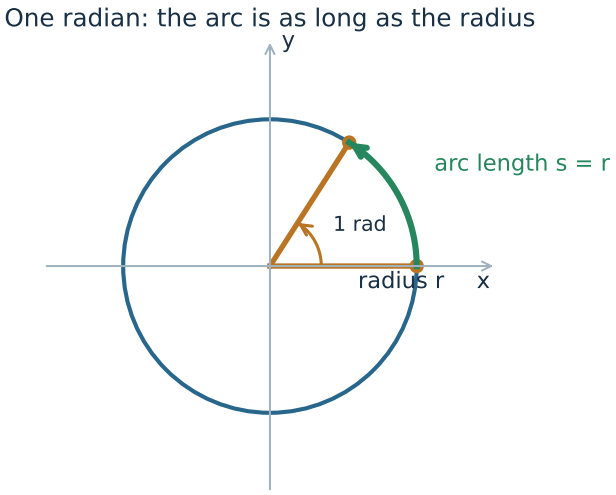
1.3 Radians compare an arc with a radius

Degrees divide a full turn into 360 equal pieces. Radians measure the same turning in a way tied directly to the circle. An *arc* is a curved part of a circle. For a central angle, the two rays meet at the circle's center and cut off an arc on its circumference.

Take the length of that arc and divide by the radius, using the same length units for both. The ratio is the angle's magnitude in radians:

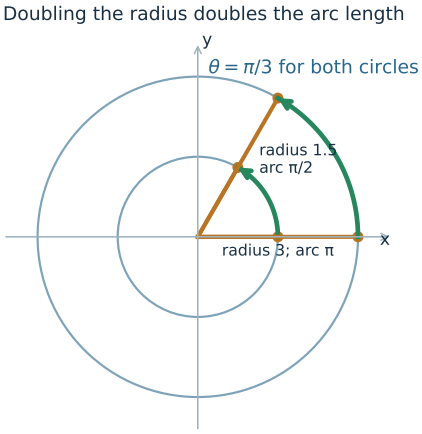
$$\boxed{\theta = \frac{s}{r}} \quad \text{(radian measure)}.$$

An angle of 1 radian cuts off an arc whose length equals the radius. It is a little less than $60°$: about $57.3°$. A radian is an angle unit; it is not another name for π.



One radian means one radius-length of arc.

Why does the size of the circle not change the angle? If you double the radius while keeping the same two rays, the arc also doubles. The ratio s/r stays the same. The picture below shows this for an angle of $\pi/3$ radians.



Angle describes turning. Radius describes size. The ratio of arc length to radius separates these ideas.

Why one turn is 2π radians

A circle's circumference is $2\pi r$. For a complete turn, divide the full circumference by the radius:

$$\theta = \frac{2\pi r}{r} = 2\pi.$$

Thus $360^\circ = 2\pi$ radians and $180^\circ = \pi$ radians. The number π is about 3.14159. A half-turn has radian measure π because its arc has length πr.

For clockwise motion, use a negative angle. If s means ordinary arc length, it is nonnegative and s/r gives the angle's magnitude. A signed angle requires an orientation as well. We will keep signed arc length and distance traveled separate.

Check 3

An arc is 9 cm long on a circle of radius 6 cm. What is the angle's magnitude in radians? At the same angle on a circle of radius 12 cm, how long is the arc?

1.4 Convert units without changing the angle

The relation $180^\circ = \pi$ radians lets us write the same angle in either unit. To convert, multiply by a fraction that equals 1 and cancels the unwanted unit.

$$\text{Degrees to radians: multiply by } \frac{\pi \text{ rad}}{180^\circ}.$$

$$\text{Radians to degrees: multiply by } \frac{180^\circ}{\pi \text{ rad}}.$$

Degrees to radians: 240°

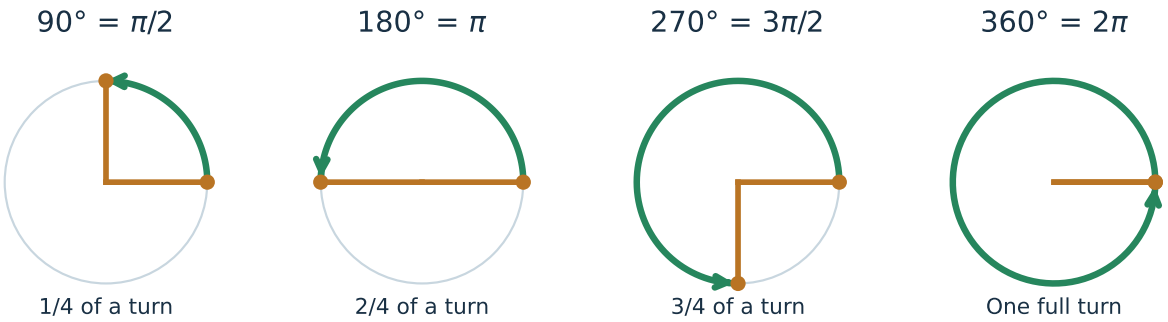
$$240^\circ \cdot \frac{\pi \text{ rad}}{180^\circ} = \frac{240\pi}{180} \text{ rad} = \frac{4\pi}{3} \text{ rad}.$$

The degree units cancel. Before calculating, we know that 240° is between a half-turn and a full turn. Our result, $4\pi/3$, is between π and 2π, as expected.

Radians to degrees: -5π/3

$$\left(-\frac{5\pi}{3} \text{ rad}\right) \frac{180^\circ}{\pi \text{ rad}} = -\frac{5(180^\circ)}{3} = -300^\circ.$$

The π factors and radian units cancel. Keep the negative sign: changing units does not change the direction of rotation.



A fraction of a full turn is the same fraction of 360° or of 2π radians.

Degrees	Radians	Fraction of one turn
30°	$\pi/6$	1/12
45°	$\pi/4$	1/8
60°	$\pi/3$	1/6
90°	$\pi/2$	1/4
180°	π	1/2
270°	$3\pi/2$	3/4
360°	2π	1

Leave an answer such as $4\pi/3$ exact unless a decimal is requested. To estimate it, use the calculator's π key and round at the end. Replacing π by 3.14 too early can add avoidable error.

Check the calculator's angle mode

In degree mode, $\sin(30) = 1/2$. In radian mode, the same turn is entered as $\sin(\pi/6) = 1/2$. Typing 30 in radian mode means 30 radians, a very different angle. Converting an angle and changing the calculator mode are separate actions.

Check 4

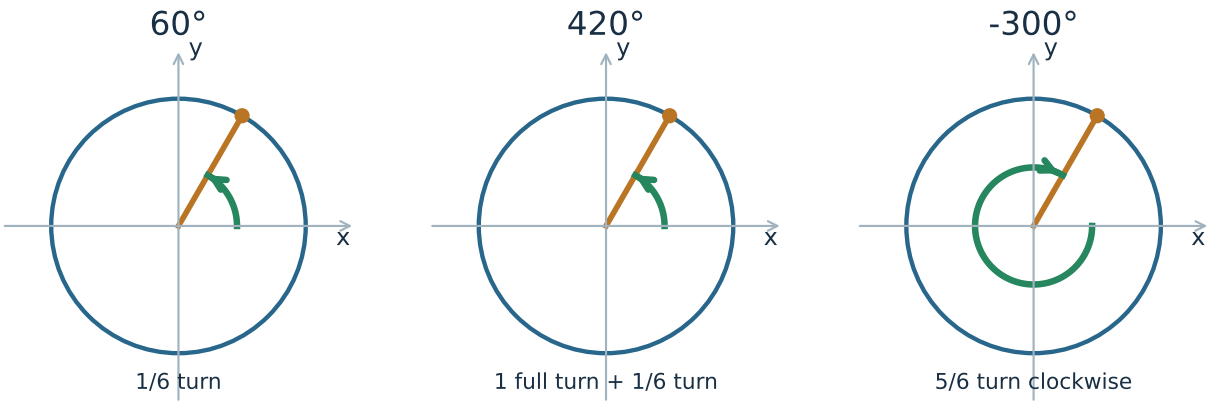
Convert $72°$ to radians and $7\pi/5$ radians to degrees. Check each result against one full turn.

1.5 Same final direction, different journeys

Two angles are *coterminal* when they have the same starting and terminal rays. Adding or subtracting a full turn changes the amount of rotation without changing the final direction.

$$\theta + 360°k \quad \text{or} \quad \theta + 2\pi k, \qquad k \in \mathbb{Z}.$$

The symbol $\mathbb{Z}$ means the integers: $\ldots, -2, -1, 0, 1, 2, \ldots$. Choose the formula that matches your angle unit. For example, $60°$, $420°$, and $-300°$ all end on the same ray.



A terminal ray does not tell us how many turns happened along the way.

Find a representative between 0 and 2π

Start with $-13\pi/4$. A full turn is $2\pi = 8\pi/4$, so add full turns using a common denominator:

$$-\frac{13\pi}{4} + \frac{8\pi}{4} = -\frac{5\pi}{4}, \qquad -\frac{5\pi}{4} + \frac{8\pi}{4} = \frac{3\pi}{4}.$$

Now $0 \le 3\pi/4 < 2\pi$. The terminal ray is in the upper-left part of the plane. We added two full turns; the original clockwise rotation was not erased from its physical history.

The interval $[0, 2\pi)$ includes 0 and excludes 2π. Those two angles point in the same direction, so

including both would repeat a representative. Similarly, use $[0°, 360°)$ for a single representative in degrees.

For the final point on a circle, a coterminal angle is enough. For distance traveled, use the actual rotation. A marker turning through 420° travels seven times as far as one turning through 60° on the same circle.

Check 5

Find the angle in $[0°, 360°)$ coterminal with 870°. How many full turns did you set aside?

1.6 Arc length, distance, and displacement

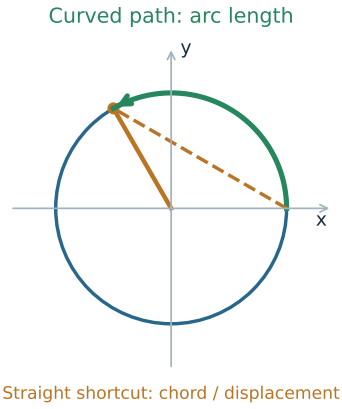
The equation $\theta = s/r$ can be rearranged to find arc length. When θ is a signed angle in radians, $r\theta$ gives a *directed* arc length:

$$s_{\text{directed}} = r\theta.$$

If the motion proceeds in one direction without reversing, the distance traveled is

$$\boxed{\text{distance} = r|\theta|}.$$

The absolute-value bars remove the sign. A clockwise journey still covers a positive distance. If motion reverses, calculate the distance for each part and add those distances. The final net angle alone will not tell you the whole distance.



Distance along the rim follows the arc. Displacement compares the starting and ending positions directly.

A clockwise turn on a radius-6 circle

A point turns clockwise through $120°$ on a circle of radius 6 cm. Convert first:

$$\theta = -120° \frac{\pi}{180°} = -\frac{2\pi}{3} \text{ rad.}$$

Then the directed arc is $6(-2\pi/3) = -4\pi$ cm, while the distance traveled is 4π cm, about 12.57 cm. The negative directed arc records clockwise orientation; it is not a negative distance.

Why radians are required

For a quarter-turn, $\theta = \pi/2$, so the arc is $r\pi/2$, one quarter of $2\pi r$. Putting 90 directly into $r\theta$ would give $90r$, which is much larger than the whole circumference. Convert degrees before multiplying.

Displacement is the change in position from the start to the finish. Its magnitude is the straight-line distance between the endpoints, not the arc length. After one complete revolution, a point has traveled a whole circumference but has zero displacement.

Check 6

A point on a circle of radius 2 m moves clockwise through half a turn. Find the signed angle, distance traveled, and magnitude of displacement.

1.7 Angular velocity and tangential speed

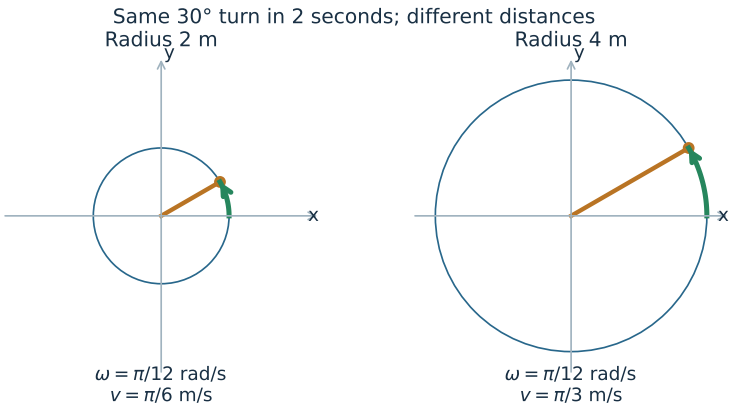
Angular velocity describes the rate at which the angle changes. Over a time interval of length Δt, the average angular velocity is

$$\omega = \frac{\Delta\theta}{\Delta t}.$$

The Greek letter ω is "omega," and Δ means "change in." With angles in radians and time in seconds, the units are radians per second, written rad/s. Counterclockwise angular velocity is positive; clockwise angular velocity is negative. Angular *speed* is the nonnegative magnitude $|\omega|$.

For uniform rotation, the rate is constant. A point farther from the center travels a longer arc during the same turn, so its tangential speed is greater:

$$\boxed{v = r|\omega|}.$$



The angular speed is shared by the whole wheel. Tangential speed depends on distance from the axis.

Convert revolutions per minute

A disk makes 6 counterclockwise revolutions per minute. A marker sits 0.5 m from the axis. Convert the angular rate one unit at a time:

$$6\frac{\text{rev}}{\text{min}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = \frac{\pi}{5} \text{ rad/s}.$$

The revolutions and minutes cancel. The marker's tangential speed is $v = 0.5(\pi/5) = \pi/10$ m/s, about 0.314 m/s. A marker twice as far from the axis would move twice as fast.

The *period* T is the time needed for one complete revolution. For uniform rotation, angular speed is $2\pi/T$. If the rotation is counterclockwise, $\omega = 2\pi/T$; if clockwise, $\omega = -2\pi/T$. A smaller period means a faster rotation.

Average speed and instantaneous speed agree here because the rotation rate is constant. If the wheel speeds up and slows down, one average rate describes only the chosen interval.

Check 7

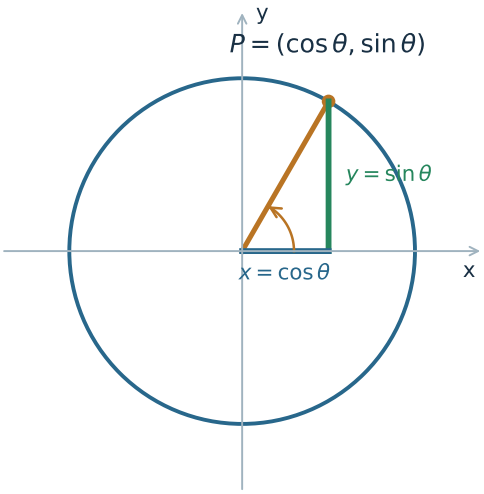
A wheel of radius 2 m makes one counterclockwise revolution every 20 seconds. Find its angular speed and tangential speed.

1.8 Sine and cosine are coordinates

The *unit circle* has center $(0, 0)$ and radius 1. Start at $(1, 0)$ and turn through angle θ. The coordinates of the resulting point define two functions:

$$\boxed{P = (\cos\theta, \sin\theta)}.$$

Cosine is the horizontal coordinate. Sine is the vertical coordinate. The expression $\sin\theta$ means "the sine of the angle θ," not "sin multiplied by theta." A function takes an input and assigns an output; here the input is an angle and the output is a coordinate.



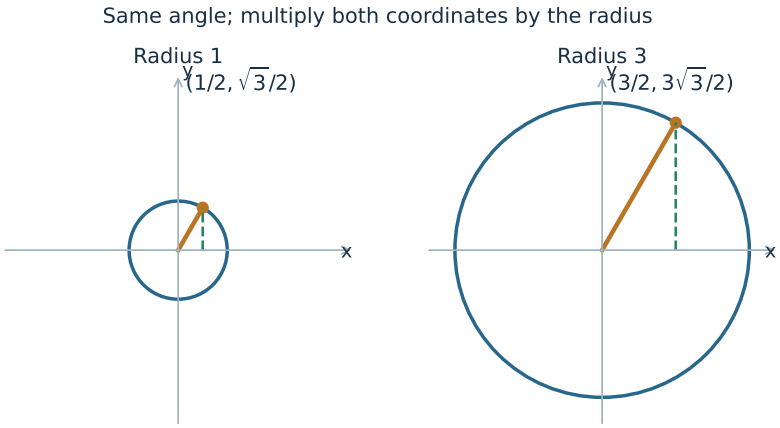
Keep the coordinate order: cosine first, sine second.

The familiar right-triangle ratios are built into this definition. In the first quadrant, the triangle has hypotenuse 1, adjacent side x, and opposite side y. Thus $\cos\theta = x/1 = x$ and $\sin\theta = y/1 = y$. The circle definition also handles angles beyond a right triangle, including negative angles and multiple turns.

Since every unit-circle point satisfies $x^2 + y^2 = 1$, its coordinates satisfy

$$\boxed{\cos^2\theta + \sin^2\theta = 1}.$$

The notation $\cos^2\theta$ means $(\cos\theta)^2$. Both coordinates lie between -1 and 1. These are useful checks on calculated values.



Changing the radius changes the physical coordinates, while sine and cosine of the angle stay the same.

Scale from a unit circle to a real wheel

If a point at angle θ has unit-circle coordinates $(3/5, 4/5)$, a point at that angle on a radius-5 circle has coordinates $(3, 4)$. In general, on a circle centered at the origin:

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

These are physical coordinates. Dividing them by the radius recovers the unit-circle coordinates. The radius does not change the angle's sine or cosine.

Check 8

The unit-circle point at an angle is $(-3/5, -4/5)$. What is the corresponding point on a circle of radius 10 centered at the origin?

1.9 Derive exact values from two triangles

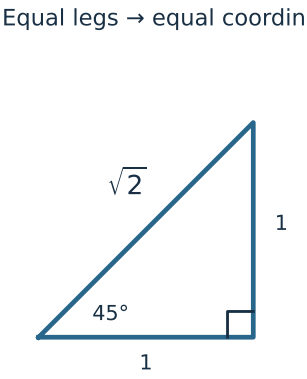
You do not need a separate rule for every special angle. Two right triangles supply the coordinate magnitudes for 30°, 45°, and 60°. The Pythagorean theorem explains where the square roots come from.

The 45° triangle

A right triangle with equal legs has two equal acute angles, each 45°. If each leg is 1, its hypotenuse is $\sqrt{1^2 + 1^2} = \sqrt{2}$. Divide every side by $\sqrt{2}$ to make the hypotenuse 1:

$$\cos 45^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

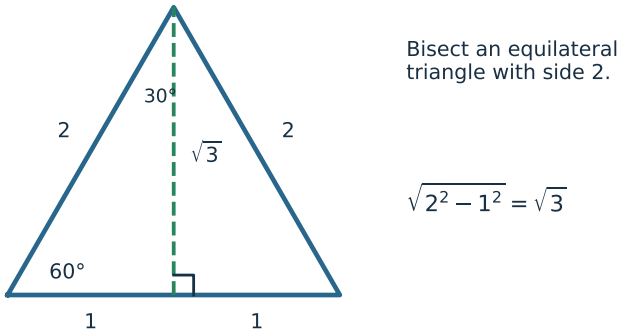
The last equality comes from multiplying top and bottom by $\sqrt{2}$. It changes the form of the number, not its value.



Equal legs give equal horizontal and vertical coordinates after normalization.

The 30° and 60° triangles

Cut an equilateral triangle of side 2 into two equal right triangles. Each has hypotenuse 2, short leg 1, and long leg $\sqrt{2^2 - 1^2} = \sqrt{3}$. The angles are 30°, 60°, and 90°.



Halving an equilateral triangle produces the ratio $1 : \sqrt{3} : 2$.

At 30°, the horizontal leg is $\sqrt{3}$ and the vertical leg is 1. Divide by the hypotenuse 2 to get $(\sqrt{3}/2, 1/2)$. At 60°, the roles of the two legs switch.

Angle	Radians	Cosine: horizontal	Sine: vertical
0°	0	1	0
30°	$\pi/6$	$\sqrt{3}/2$	$1/2$
45°	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$
60°	$\pi/3$	$1/2$	$\sqrt{3}/2$
90°	$\pi/2$	0	1

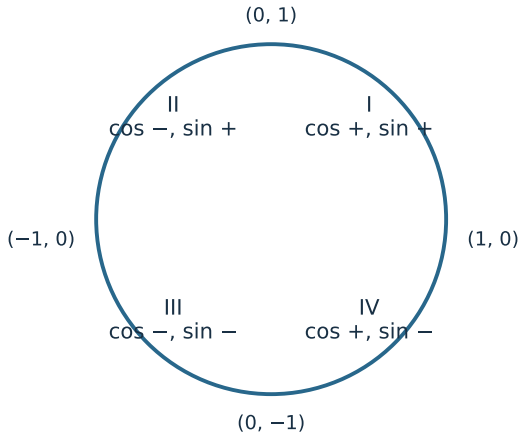
The axis points at 0° and 90° come directly from the circle. They do not require a right triangle with a zero-length leg.

Check 9

Why is $\sin 60°$ greater than $\sin 30°$? Check that the two coordinates at 45° satisfy the unit-circle equation.

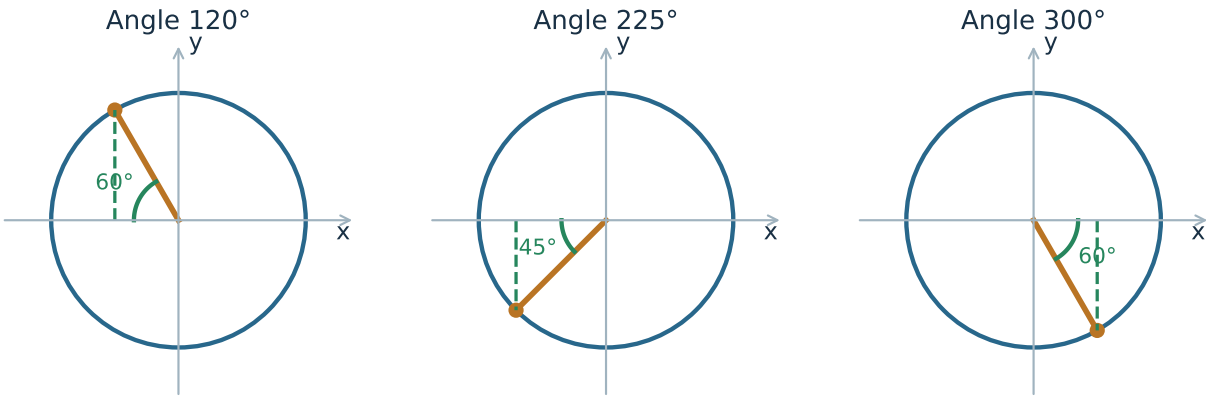
1.10 Use reference angles and quadrant signs

The axes divide the plane into four open regions called *quadrants*. Number them counterclockwise, beginning with the upper-right region. A point on an axis belongs to none of the open quadrants.



The signs follow from left/right and above/below. They are coordinate signs, not an extra rule to memorize.

For a terminal ray away from the axes, the *reference angle* is the acute angle between that ray and the horizontal axis. "Acute" means greater than $0°$ and less than $90°$. Always measure it to the horizontal axis, which may mean the negative x-axis.



Use the reference triangle for magnitudes, then the quadrant for signs.

Find the point at 120°

First locate the angle: $120°$ is in Quadrant II. The reference angle is $180° - 120° = 60°$. At $60°$, the coordinate magnitudes are $1/2$ and $\sqrt{3}/2$. Quadrant II lies left and up, so

$$(\cos 120°, \sin 120°) = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

The horizontal coordinate gets the minus sign. The vertical coordinate remains positive. Squaring and adding gives $1/4 + 3/4 = 1$, as it should.

When the original angle includes several turns, first find a coterminal angle in one revolution. Then locate its quadrant and reference angle. For a point on an axis, read the coordinates directly: $(1, 0)$, $(0, 1)$, $(-1, 0)$, or $(0, -1)$. No acute reference triangle is needed.

A useful routine is: locate the ray, find the acute reference angle, determine the two magnitudes, and assign each sign from the picture. Keep the order $(\cos\theta, \sin\theta)$.

Check 10

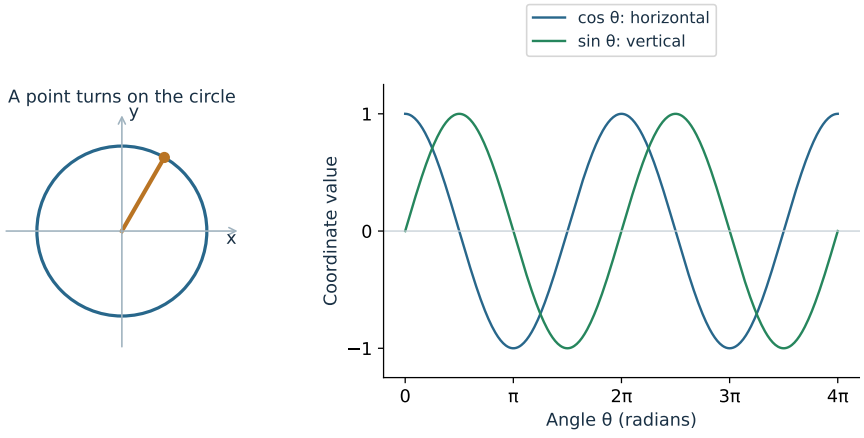
Find the unit-circle point at 300°. Does −60° give the same point? Explain.

1.11 A moving coordinate traces a wave

A point turning around a circle moves in two dimensions. We can also study one coordinate at a time. Make a graph whose horizontal axis is the angle θ, and whose vertical axis is the point's vertical coordinate. That graph is $y = \sin\theta$.

Be careful about the two pictures. On the circle, x and y describe physical position. On the graph of sine, the horizontal axis is the *angle input*; it is not the point's horizontal position.

Angle	Point on the circle	Cosine	Sine
0	$(1, 0)$	1	0
$\pi/2$	$(0, 1)$	0	1
π	$(-1, 0)$	-1	0
$3\pi/2$	$(0, -1)$	0	-1
2π	$(1, 0)$	1	0



One full turn produces one complete cycle of each coordinate graph.

Between these quarter turns, the point moves smoothly, so its coordinates change smoothly. Sine rises from 0 to 1, falls through 0 to -1, and returns to 0. Cosine begins at 1 because the starting point is $(1, 0)$.

A function is *periodic* if its values repeat after a fixed change in input. Adding a complete turn returns the point to exactly the same position:

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

The period of each is 2π. A half-turn is generally not a period: it moves a point to the opposite side of the circle and reverses both coordinates. For instance, cosine is 1 at 0 and -1 at π.

Explore the connection

In Circle to Wave, stop at each quarter-turn and compare the point's height with the sine graph. Then show two full turns. The playback speed changes how quickly the demonstration runs; the graph against angle keeps the same shape. A graph against time requires a rotation rate, which we introduce next.

Check 11

Why does sine start at 0 while cosine starts at 1? How many complete cycles appear from 0 to 4π?

1.12 Build a model of height over time

To describe a turning wheel over time, connect two steps: time determines an angle, and the angle determines a position. For uniform rotation starting at angle zero,

$$\theta(t) = \omega t.$$

Here t is elapsed time. The notation $\theta(t)$ means the angle at time t. If ω is in rad/s and t is in seconds, their product is in radians.

For a wheel of radius r, centered horizontally at 0 and vertically at height c, with a rider starting at the rightmost point,

$$\boxed{x(t) = r\cos(\omega t), \qquad h(t) = c + r\sin(\omega t)}.$$

The sine term is vertical displacement from the center. Adding c gives height above ground. The radius stretches the unit-circle coordinate; it does not change the value of sine itself.

A radius-8 Ferris wheel

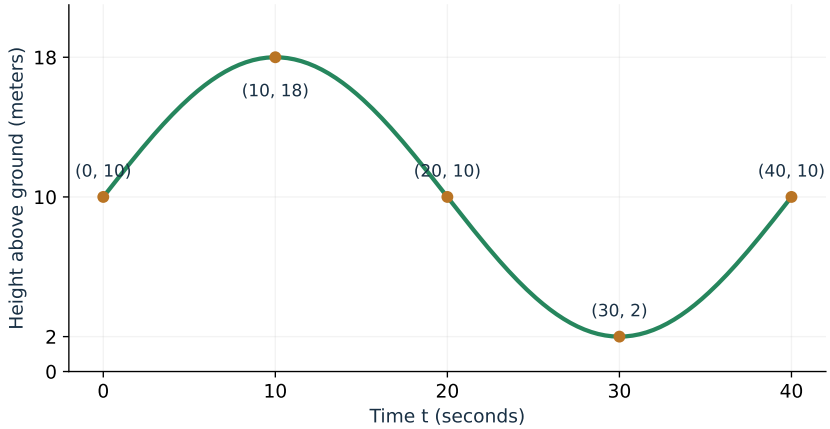
A wheel has radius 8 m, center height 10 m, and period 40 s. The rider starts at the rightmost point, and the wheel turns counterclockwise. First find the angular velocity:

$$\omega = \frac{2\pi}{40} = \frac{\pi}{20} \text{ rad/s}.$$

Next substitute the radius, center height, and angular rate:

$$x(t) = 8\cos\left(\frac{\pi t}{20}\right), \qquad h(t) = 10 + 8\sin\left(\frac{\pi t}{20}\right).$$

At $t = 10$, the angle is $\pi/2$, so the rider is at $(0, 18)$. This is the top: the 10 m center height plus the 8 m radius. At $t = 30$, the angle is $3\pi/2$, so the height is $10 - 8 = 2$ m.



The horizontal axis now measures seconds. The physical assumptions determine where the cycle begins and how long it takes.

Check a model against the situation before trusting a calculated value. The initial height should be 10 m, the height should stay between 2 and 18 m, and one full cycle should take 40 seconds. A negative height would signal a problem with this physical setup.

If a rider begins at another angle θ_0, use $\theta(t) = \theta_0 + \omega t$ inside the coordinate formulas. For clockwise rotation, use a negative ω. Starting position and direction matter; the radius and period alone do not determine a unique height formula.

This is the connection we will develop further: circular motion gives coordinate functions, and those functions let us model repeating behavior.

Check 12

In the example, what is the height at 20 seconds? If the radius became 12 m but the center stayed 10 m high, what would the model predict at the bottom?

1.13 Choose a relationship

Question	Relationship	What to check
How much turning?	$180^\circ = \pi$ rad	Keep the sign; match the unit.
Same terminal ray?	$\theta + 2\pi k$, $k \in \mathbb{Z}$	Use full turns, not half-turns.
How far along the rim?	$d = r\lvert\theta\rvert$	Radians; one direction without reversing.
How fast is the turning?	$\omega = \Delta\theta/\Delta t$	Signed rate, such as rad/s.
How fast is the rim point?	$v = r\lvert\omega\rvert$	Distance/time, such as m/s.
Where is the unit-circle point?	$(\cos\theta, \sin\theta)$	Horizontal first; check signs.
Where is the radius-r point?	$(r\cos\theta, r\sin\theta)$	Center at the origin.
Height on a wheel?	$h = c + r\sin(\omega t)$	Rightmost start, constant rotation rate.

1.14 Practice with explanation

Attempt each problem before consulting the explanations.

Try it. Sketch -120°, 240°, and 600°. Which terminal rays coincide? Justify the result using coterminal angles.

Try it. Convert 225° to radians and $-\pi/3$ radians to degrees, and sketch their terminal rays.

Try it. A Ferris wheel of radius 12 m makes one revolution every 40 s. Find its angular speed and a passenger's tangential speed.

Try it. A wheel of radius 0.6 m turns through 135° in 3 seconds. Find the arc length, average angular speed, and average tangential speed. Explain the units of all three answers.

Try it. Without a calculator, describe the point on the unit circle at 150° and at 330°. Which coordinates have the same absolute value? Which signs differ?

Try it. A different wheel has radius 6 meters, center height 7 meters, and period 30 seconds. Starting from the rightmost point, write its coordinate model and find the passenger's height after 7.5 seconds.

1.15 Review exercises

1. Convert -135° and 300° to radians.
2. Convert $11\pi/6$ and $-5\pi/4$ to degrees.
3. Find an angle in $[0, 2\pi)$ coterminal with $19\pi/6$.
4. A circle of radius 8 cm subtends an angle of $5\pi/12$. Find the arc length.
5. A turntable makes $33\frac{1}{3}$ revolutions per minute. Find angular speed in radians per second.
6. Explain why sine and cosine must lie between -1 and 1.
7. Find unit-circle coordinates at $2\pi/3$, $5\pi/4$, and $11\pi/6$.
8. A Ferris wheel has radius 10 m and center $(0, 12)$. Write a rider's coordinates after turning through θ from the rightmost point.
9. A point begins at $(1, 0)$ and turns through $-7\pi/6$ radians. Determine a coterminal angle in $[0, 2\pi)$ and give the exact point.
10. A wheel completes 9 revolutions in 2 minutes. Determine its angular speed in both radians per second and degrees per second.
11. A rider on a circle of radius 4 m moves through 75°. Find the arc length exactly in terms of π and approximately to two decimals.
12. Describe the coordinates at $3\pi/2$ and explain why a terminal point on an axis does not belong to any open quadrant.
13. For the Ferris-wheel model in the chapter, explain which physical quantity would change if the coefficient 8 in both coordinate equations became 12.

1.16 Explanations for the checks

1. The distance is $\sqrt{(-3)^2 + 4^2} = 5$. The normalized point is $(-3/5, 4/5)$. It stays left of the vertical axis and above the horizontal axis.

2. $-90°$ is one quarter-turn clockwise and ends on the negative y-axis. $450°$ is one full turn plus a quarter-turn counterclockwise and ends on the positive y-axis.

3. The angle is $9/6 = 1.5$ radians. Doubling the radius doubles the arc, so the new arc is 18 cm. The angle remains 1.5 radians.

4. $72(\pi/180) = 2\pi/5$ radians, one fifth of a turn. $(7\pi/5)(180/\pi) = 252°$, seven tenths of a turn.

5. $870° - 2(360°) = 150°$. We set aside two full turns. The original rotation was two turns plus $150°$.

6. The angle is $-\pi$ radians. Distance is $2|-\pi| = 2\pi$ m. The endpoints are opposite ends of a diameter, so the displacement has magnitude 4 m.

7. Angular speed is $2\pi/20 = \pi/10$ rad/s. Tangential speed is $2(\pi/10) = \pi/5$ m/s. The units distinguish turning rate from distance rate.

8. Multiply both coordinates by 10: $(-6, -8)$. Its distance from the origin is $\sqrt{36 + 64} = 10$, and both signs remain negative.

9. The point at $60°$ is higher: $\sqrt{3}/2 > 1/2$. At $45°$, $(\sqrt{2}/2)^2 + (\sqrt{2}/2)^2 = 1/2 + 1/2 = 1$.

10. The reference angle is $60°$, and Quadrant IV is right and down. The point is $(1/2, -\sqrt{3}/2)$. Since $-60° + 360° = 300°$, the angles are coterminal and give the same point.

11. The starting point is $(1, 0)$, so the horizontal coordinate is 1 and the vertical coordinate is 0. The interval from 0 to 4π contains two full turns and therefore two cycles.

12. At 20 seconds, $\theta = \pi$, so $h = 10$ m. A radius of 12 m with the same center would predict a bottom height of $10 - 12 = -2$ m. A wheel meant to stay above ground would need a higher center or a different physical setup.

1.17 Practice solutions

1. All three terminal rays coincide. $-120° + 360° = 240°$, and $600° - 360° = 240°$. They point down and left, although the signed rotations differ.

2. $225°(\pi/180°) = 5\pi/4$ radians, in Quadrant III. $(-\pi/3)(180°/\pi) = -60°$, in Quadrant IV.

3. One revolution is 2π radians, so angular speed is $2\pi/40 = \pi/20$ rad/s. Tangential speed is $12(\pi/20) = 3\pi/5$ m/s, about 1.885 m/s.

4. Convert $135° = 3\pi/4$ radians. Arc length is $0.6(3\pi/4) = 9\pi/20$ m. Average angular speed is $(3\pi/4)/3 = \pi/4$ rad/s. Average tangential speed is $(9\pi/20)/3 = 3\pi/20$ m/s, assuming the turn proceeds in one direction.

5. At $150°$, the point is $(-\sqrt{3}/2, 1/2)$. At $330°$, it is $(\sqrt{3}/2, -1/2)$. Corresponding coordinates have equal absolute values, and both signs are opposite.

6. The direction is not specified, so state an assumption. For counterclockwise rotation, $\omega = 2\pi/30 = \pi/15$, $x(t) = 6\cos(\pi t/15)$, and $h(t) = 7 + 6\sin(\pi t/15)$. At 7.5 s the angle is $\pi/2$, so the height is 13 m. Clockwise rotation would instead put the rider at height 1 m at that time.

1.18 Review solutions

1. $-135° = -3\pi/4$ radians; $300° = 5\pi/3$ radians. Multiply each degree value by $\pi/180$.

2. $11\pi/6 = 330°$; $-5\pi/4 = -225°$. Multiply each radian value by $180/\pi$.

3. $19\pi/6 - 2\pi = 19\pi/6 - 12\pi/6 = 7\pi/6$, which is in $[0, 2\pi)$.

4. $s = 8(5\pi/12) = 10\pi/3$ cm. The angle is already in radians.

5. $33\frac{1}{3} = 100/3$ rev/min. Multiply by 2π rad/rev and divide by 60 s/min to get $10\pi/9$ rad/s.

6. They are coordinates on a radius-1 circle centered at the origin. No such point can be farther than 1 unit horizontally or vertically from the origin.

7. At $2\pi/3$: $(-1/2, \sqrt{3}/2)$. At $5\pi/4$: $(-\sqrt{2}/2, -\sqrt{2}/2)$. At $11\pi/6$: $(\sqrt{3}/2, -1/2)$. Use the reference angle for magnitudes and the quadrant for signs.

8. $x = 10\cos\theta$, $y = 12 + 10\sin\theta$. The center adds 12 to the vertical coordinate; the radius multiplies both unit-circle coordinates.

9. $-7\pi/6 + 2\pi = 5\pi/6$. The point is $(-\sqrt{3}/2, 1/2)$, in Quadrant II.

10. In 120 seconds the wheel turns through 18π radians, or 3240°. Divide each by 120: the angular speed is $3\pi/20$ rad/s, or 27°/s.

11. $75° = 5\pi/12$ radians, so the arc is $4(5\pi/12) = 5\pi/3$ m, approximately 5.24 m.

12. The point is $(0, -1)$, on the negative y-axis. Open quadrants exclude their boundary axes, so it belongs to none of them.

13. The radius would increase from 8 to 12 m, increasing the amplitude of each coordinate function to 12. The period remains 40 s and the center height remains 10 m. The predicted height range becomes $[-2, 22]$ m; a wheel staying above ground would require a higher center.

2 The Six Trigonometric Functions

A point on a rotating wheel has a horizontal position and a vertical position. Last module we named those coordinates cosine and sine. This module asks what else the same point can tell us: the slope of its direction, the ratios of triangle sides, and the places where a division stops making sense.

The six trigonometric functions come from these connected measurements. Begin with a point or a triangle, then form the ratio the question needs.

2.1 Build six functions from two coordinates

Let $P = (x, y)$ be the point at angle θ on the unit circle. Its distance from the origin is 1, so

$$x = \cos\theta, \qquad y = \sin\theta.$$

The other four functions use these coordinates:

$$\tan\theta = \frac{y}{x}, \quad \cot\theta = \frac{x}{y},$$
$$\sec\theta = \frac{1}{x}, \quad \csc\theta = \frac{1}{y}.$$

Each quotient requires a nonzero denominator. These definitions work in every quadrant and on any coordinate axis where the needed denominator is nonzero.

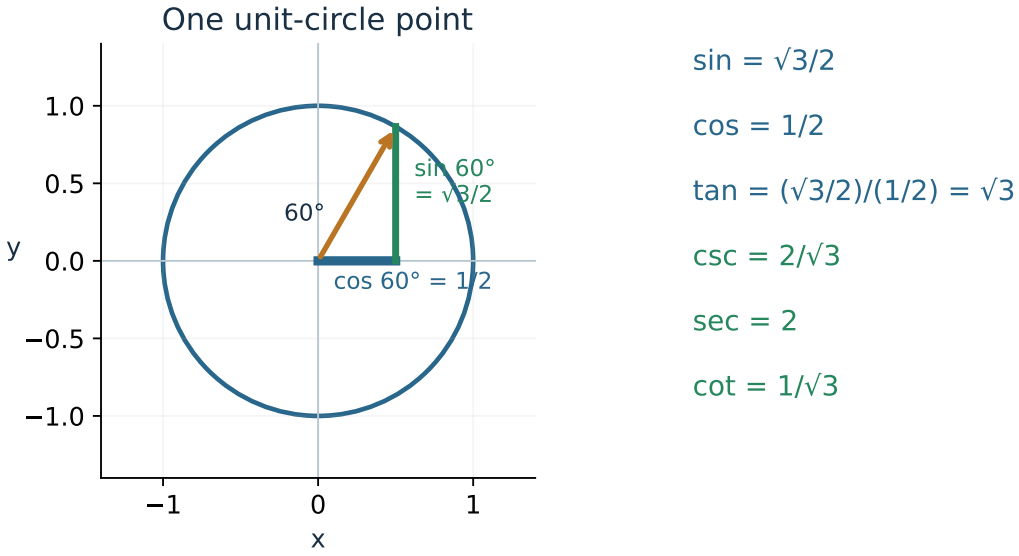


Figure 1: The horizontal and vertical coordinates determine all the ratios at a 60-degree angle.

Worked example: six values at one angle. At $\theta = \pi/3 = 60°$, the point is $(1/2, \sqrt{3}/2)$. Therefore

$$\sin\theta = \frac{\sqrt{3}}{2}, \quad \cos\theta = \frac{1}{2}, \quad \tan\theta = \frac{\sqrt{3}/2}{1/2} = \sqrt{3},$$
$$\csc\theta = \frac{2}{\sqrt{3}}, \quad \sec\theta = 2, \quad \cot\theta = \frac{1}{\sqrt{3}}.$$

The exact forms $2/\sqrt{3} = 2\sqrt{3}/3$ and $1/\sqrt{3} = \sqrt{3}/3$ are equivalent. Rationalizing a denominator changes the form, while preserving the value.

Secant and cosecant are the reciprocals of cosine and sine. Cotangent equals $1/\tan\theta$ whenever both expressions are defined. For its full domain, use $\cot\theta = \cos\theta/\sin\theta$: at 90°, cotangent is zero even though tangent is undefined.

Use the Six-Function Explorer to compare the values with the point's coordinates. Predict the signs before reading the outputs.

2.2 Tangent measures the slope of a direction

For a line through the origin and $P = (\cos\theta, \sin\theta)$, slope is rise divided by run:

$$m = \frac{\sin\theta}{\cos\theta} = \tan\theta.$$

The same line meets the vertical line $x = 1$ at $(1, \tan\theta)$, provided cosine is nonzero. This is a useful picture of tangent as a signed height.

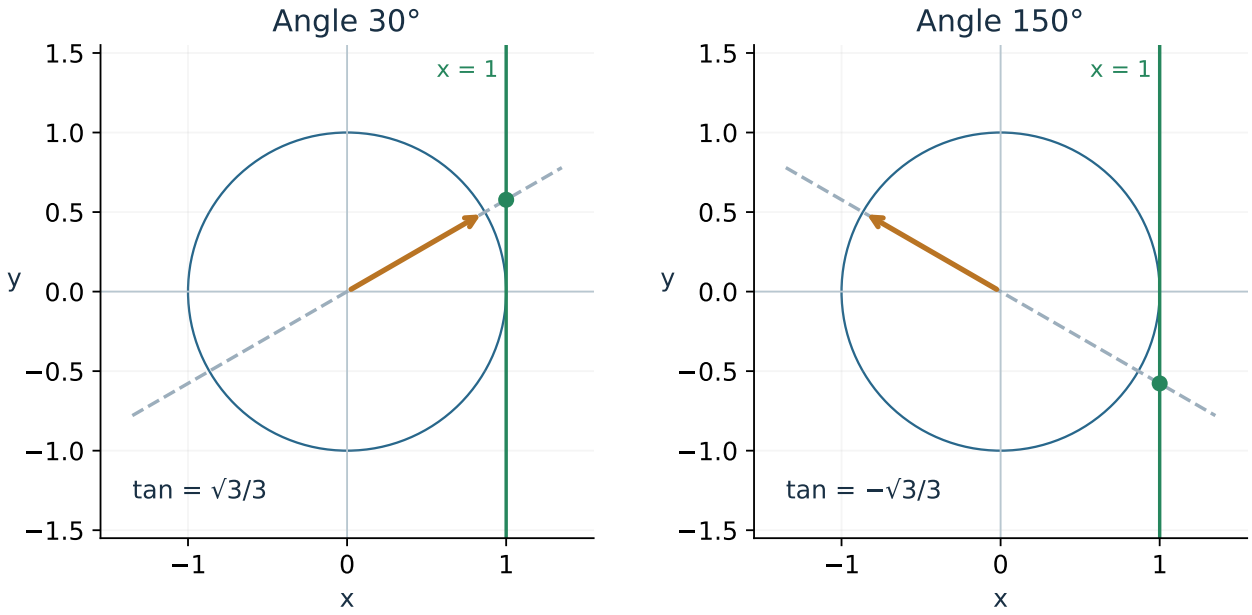


Figure 2: The line through the rotating point meets x equal to one at the height tan theta. When the original point lies on the left, extend its line through the origin.

In Quadrants II and III, the terminal ray points left, so we extend its line backward to reach $x = 1$. Keeping the whole line visible explains the tangent construction in those quadrants.

At 30°, the slope is $(1/2)/(\sqrt{3}/2) = 1/\sqrt{3}$. At 60°, it is $\sqrt{3}$. A direction closer to vertical has a larger positive slope in this first-quadrant comparison.

As θ approaches 90° from below, sine approaches 1 while cosine approaches zero through positive values. Dividing by that small positive number makes tangent large. At exactly 90°, the line is vertical and the quotient is undefined.

2.3 Recover the familiar right-triangle ratios

Consider a right triangle with an acute angle θ, opposite side o, adjacent leg a, and hypotenuse h. Scale every side by $1/h$. The hypotenuse becomes 1, and the other two sides become a/h and o/h, the coordinates of a unit-circle point.

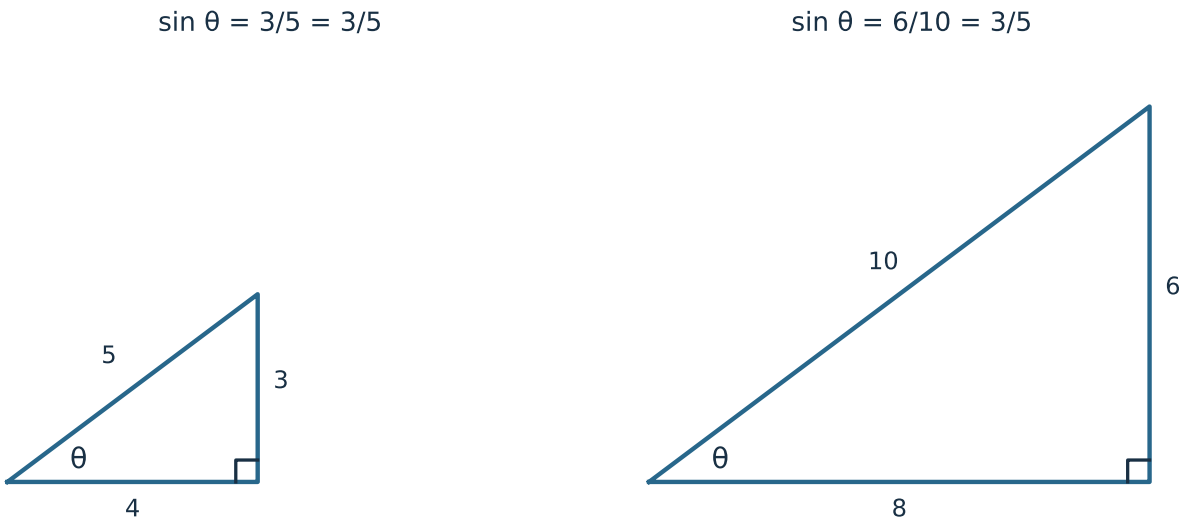


Figure 3: Similar triangles keep the same ratios even when every side length changes.

It follows that

$$\sin\theta = \frac{o}{h}, \qquad \cos\theta = \frac{a}{h}, \qquad \tan\theta = \frac{o}{a}.$$

Their reciprocal ratios are $\csc\theta = h/o$, $\sec\theta = h/a$, and $\cot\theta = a/o$. Opposite and adjacent depend on which acute angle you choose. The hypotenuse always faces the right angle.

Worked example: a support cable. A cable is 13 m long and reaches a point 12 m above level ground. Its horizontal run is $\sqrt{13^2 - 12^2} = 5$ m. For the angle with the ground,

$$\sin\theta = \frac{12}{13}, \quad \cos\theta = \frac{5}{13}, \quad \tan\theta = \frac{12}{5}.$$

The reciprocal ratios are $13/12, 13/5, 5/12$, respectively. Each compares two lengths from the same triangle.

Try it. A right triangle has legs 5 and 12. Choose the angle opposite the side of length 5. Find the hypotenuse and all six ratios.

2.4 Exact values come from two special triangles

An equilateral triangle of side 2 splits into two right triangles with sides $1, \sqrt{3}, 2$. This gives the 30° and 60° values. A right triangle with equal legs 1 has hypotenuse $\sqrt{2}$, giving the 45° values.

Angle	$\sin\theta$	$\cos\theta$	$\tan\theta$
$30^\circ = \pi/6$	$1/2$	$\sqrt{3}/2$	$\sqrt{3}/3$
$45^\circ = \pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1
$60^\circ = \pi/3$	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$

For an angle in another quadrant, use its reference angle to find the coordinate magnitudes, then assign signs from the point's location.

Worked example: a third-quadrant angle. For $5\pi/4 = 225°$, the reference angle is $\pi/4$. The point lies left of the origin and below it:

$$(\cos\theta, \sin\theta) = \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right).$$

Thus $\tan\theta = \cot\theta = 1$, while $\sec\theta = \csc\theta = -\sqrt{2}$. The two negatives cancel in the quotients.

2.5 Use signs rather than a separate rule for each function

Cosine follows the horizontal coordinate; sine follows the vertical coordinate. Their reciprocals keep those signs. Tangent and cotangent are positive when the coordinate signs agree and negative when they differ.

Quadrant	Coordinate signs (x, y)	Positive functions
I	$(+, +)$	All six
II	$(-, +)$	Sine and cosecant
III	$(-, -)$	Tangent and cotangent
IV	$(+, -)$	Cosine and secant

For $150°$, the reference angle is $30°$, so the point is $(-\sqrt{3}/2, 1/2)$. For example, $\tan 150° = -1/\sqrt{3}$ and $\csc 150° = 2$.

Try it. Suppose $\cos\theta = 3/5$, with θ in Quadrant IV. Use $x^2 + y^2 = 1$ to find sine, then find the other four functions.

2.6 Separate zero from undefined

A quotient is zero when its numerator is zero and its denominator is nonzero. It is undefined when its denominator is zero. These situations occur at different places on the circle.

At $90°$, the point is $(0, 1)$. Therefore sine is 1 and cosine is 0; tangent and secant are undefined because they divide by cosine. Cotangent is $0/1 = 0$, and cosecant is 1.

At $180°$, the point is $(-1, 0)$. Now sine is 0 and cosine is -1; cotangent and cosecant are undefined, tangent is 0, and secant is -1.

In radians, cosine vanishes at $\pi/2 + k\pi$, so tangent and secant exclude those inputs. Sine vanishes at $k\pi$, so cotangent and cosecant exclude those inputs. Here k is any integer.

Try it. Find all six values at $3\pi/2$. For each undefined value, name its denominator.

2.7 Read graphs of coordinates and reciprocals

The horizontal axis of a trig graph represents angle; its vertical axis represents a function value. This graph records a measurement from the rotating point, rather than its circular path.

Sine and cosine stay in $[-1, 1]$ and repeat after 2π. Secant and cosecant have the same full-turn periods as their reciprocal partners. Their values lie in $(-\infty, -1] \cup [1, \infty)$, since reciprocating a nonzero number of magnitude at most 1 produces a magnitude at least 1.

Near $\pi/2$, cosine approaches zero from above on the left and below on the right. Secant therefore grows positively on one side and negatively on the other. Its graph has a vertical asymptote there. The two branches remain separated.

Tangent and cotangent repeat after π. A half-turn negates both coordinates, preserving their quotient:

$$\tan(\theta + \pi) = \frac{-\sin\theta}{-\cos\theta} = \tan\theta$$

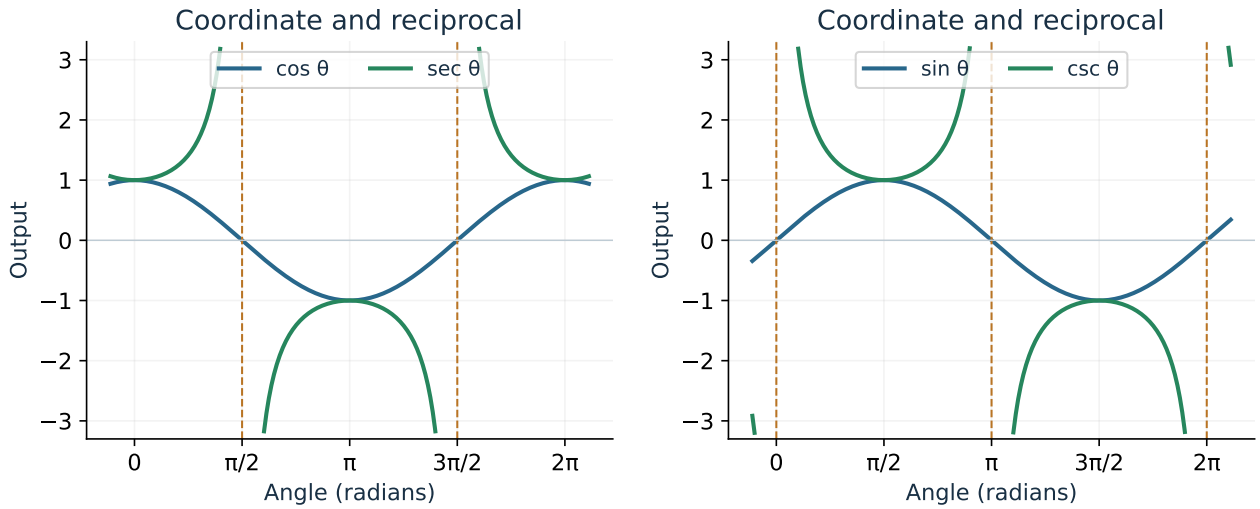


Figure 4: Cosine and its reciprocal secant share signs. Zeros of cosine become excluded inputs and vertical asymptotes of secant.

where the denominator is nonzero. Tangent has zeros at $k\pi$; cotangent has zeros at $\pi/2 + k\pi$.

Use the Six Trigonometric Graphs Explorer and the Reciprocal Graphs Explorer to compare periods, zeros, and excluded inputs. Follow the denominator when a graph breaks.

2.8 Choose a ratio for a measurement

Draw the physical situation and identify which side is known. A horizontal distance and an elevation angle suggest tangent; a line-of-sight distance and angle suggest sine for a height.

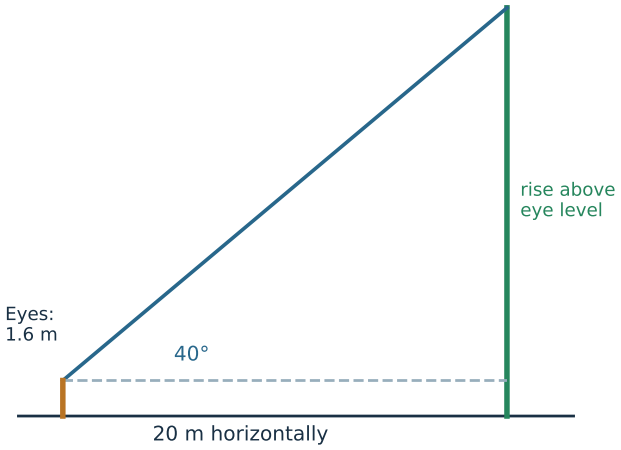


Figure 5: The height above the instrument comes from the right triangle; add instrument height to obtain total height above ground.

Worked example: a tree measured from eye level. An observer stands 20 m horizontally from a tree and measures a 40° elevation angle. The observer's eyes are 1.6 m above the level ground.

First find the rise above eye level:

$$\frac{h_{\text{rise}}}{20} = \tan 40^\circ, \qquad h_{\text{rise}} \approx 16.8 \text{ m}.$$

Adding the eye height gives a tree height of about 18.4 m. Use degree mode for this calculation and round at the end.

When the angle is unknown but a ratio is known, a calculator's inverse trig function can recover an acute angle. For a ramp rising 1.5 m over 6 m horizontally, $\theta = \arctan(1.5/6) \approx 14.0^\circ$. Module 4 will examine why inverse functions choose particular angles.

2.9 Review exercises

1. At $2\pi/3$, find all six functions exactly.
2. At $7\pi/4$, find tangent, cotangent, secant, and cosecant.
3. Explain geometrically why $\tan(\pi/2)$ is undefined.
4. For which angles in $[0, 2\pi)$ is cosecant undefined?
5. For legs 5 and 12, find all ratios for the angle opposite the side of length 5.
6. A ladder reaches 8 m high with its foot 3 m from a wall. Find the angle with the ground.
7. A drone has elevation angle 28° and horizontal distance 75 m from an observer at ground level. Estimate its altitude.
8. Divide $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$ and simplify. State the restriction.
9. Find all six exact values at $11\pi/6$.
10. Describe the zeros and asymptotes of tangent on $[-2\pi, 2\pi]$.
11. Sketch one period of secant, labeling its asymptotes and its closest values to zero.
12. Explain why neither secant nor cosecant can equal $1/2$.
13. If $\cos\theta = -12/13$ in Quadrant II, find all the other values.
14. Explain why tangent repeats after a half-turn.
15. For the point $(0, -1)$, identify its angle in $[0, 2\pi)$ and every zero or undefined function.

2.10 Check your understanding

For the 5–12 triangle, the hypotenuse is 13 and the ratios are $\sin = 5/13, \cos = 12/13, \tan = 5/12, \csc = 13/5, \sec = 13/12, \cot = 12/5$.

In the Quadrant IV problem, sine is $-4/5$, tangent $-4/3$, cosecant $-5/4$, secant $5/3$, and cotangent $-3/4$.

At $3\pi/2$, sine is -1, cosine 0, cosecant -1, and cotangent 0. Tangent and secant are undefined because cosine is zero.

3 Periodic Functions and Their Graphs

A rotating point returns to the same position after one turn. Its height also returns to the same value, producing a repeating wave when height is plotted against time. A wheel, a vibrating string, and a buoy can have very different sizes and speeds while sharing this mathematical pattern.

This chapter connects a wave's visible features to the numbers in its equation. We will read those numbers from a graph, change them one at a time, and build models from measurements.

3.1 Identify a cycle and its period

A function is periodic when a positive shift of its input reproduces all its values: $f(x+T) = f(x)$, with the domain also preserved by that shift. The least positive such T, when one exists, is the fundamental period.

For sine and cosine, a full turn restores both coordinates:

$$\sin(x + 2\pi) = \sin x, \qquad \cos(x + 2\pi) = \cos x.$$

Their fundamental period is 2π radians. If the input is time, a period has time units instead. The equation's inside angle determines how time becomes rotation.

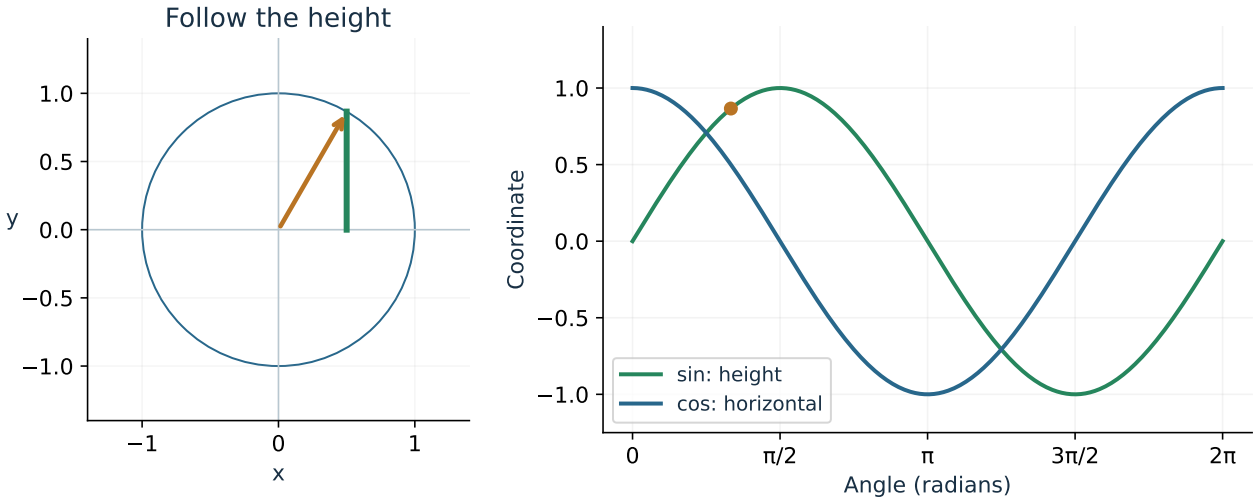


Figure 6: The same circular motion produces a sine wave from height and a cosine wave from horizontal position.

Cosine begins at 1 when its angle is zero. Sine begins at zero and initially rises. They describe coordinates a quarter-turn apart, so either can model the same repeating quantity after an appropriate shift.

3.2 Read amplitude and midline from the extremes

Consider $y = A\sin x + D$. Multiplication by A changes the vertical scale, and adding D changes the center of the wave.

The **midline** is $y = D$. The **amplitude** is the nonnegative distance $|A|$ from the midline to either extreme. Thus

$$\text{minimum} = D - |A|, \qquad \text{maximum} = D + |A|.$$

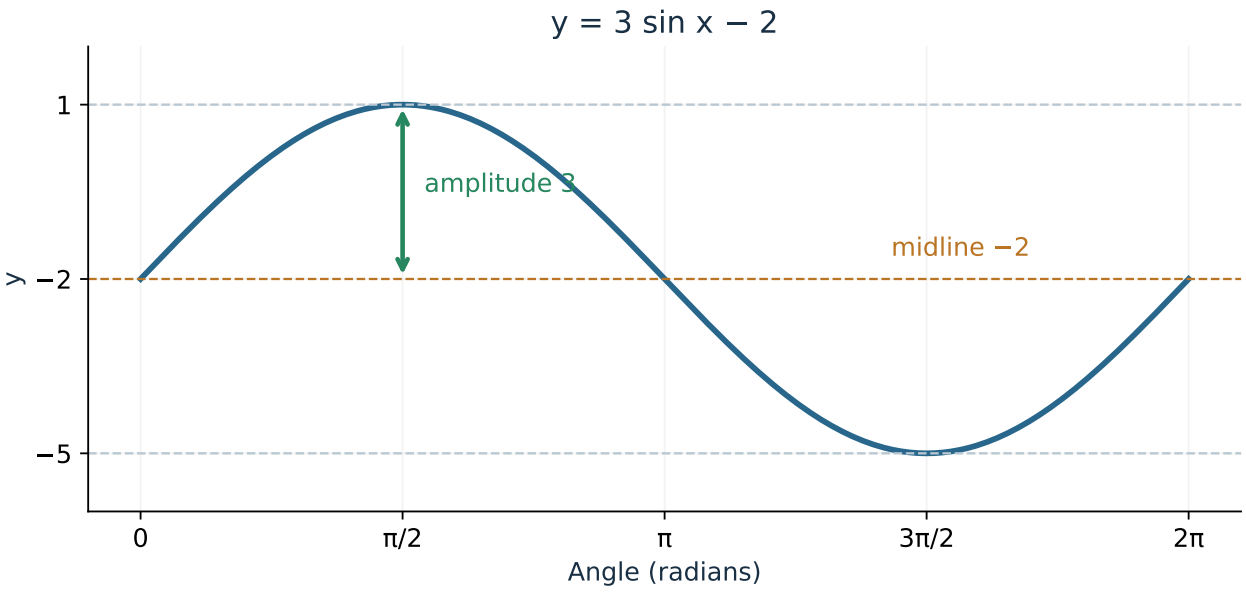


Figure 7: A transformed wave oscillates around its midline. Amplitude measures half the vertical distance between its extremes.

For $y = 3\sin x - 2$, the amplitude is 3 and the midline is -2. The minimum is -5, maximum is 1, and range is $[-5, 1]$.

If a graph or measurement gives maximum M and minimum m, recover the two quantities by

$$|A| = \frac{M - m}{2}, \qquad D = \frac{M + m}{2}.$$

The midpoint of the extremes gives the midline; half their separation gives the amplitude.

Worked example: a daily temperature model. Suppose a roughly sinusoidal temperature varies between 14°C and 26°C. The midline is 20°C, and the amplitude is 6°C. These two measurements establish the vertical scale but leave the period and starting phase to be determined.

A negative coefficient reflects a wave across its midline. The graphs $4\sin x$ and $-4\sin x$ have the same range and amplitude, but one initially rises while the other initially falls.

Try it. A signal ranges from -7 to 5. Find its amplitude and midline, then write one sine model with those extremes.

3.3 A multiplier inside the function changes the period

For $y = \sin(Bx)$, one cycle occurs when the inside angle Bx changes by 2π. Therefore

$$T = \frac{2\pi}{|B|}, \qquad B \neq 0.$$

For $\sin(2x)$, an input increase of π makes the inside angle increase by 2π. Two cycles fit where $\sin x$ completes one.

The period of $\cos(4x)$ is $\pi/2$. If x is measured in seconds and B is an angular frequency, the period is in seconds. For a model $\sin(\omega t)$,

$$\omega = \frac{2\pi}{T} \text{ radians per unit time}, \qquad f = \frac{1}{T} = \frac{|\omega|}{2\pi} \text{ cycles per unit time}.$$

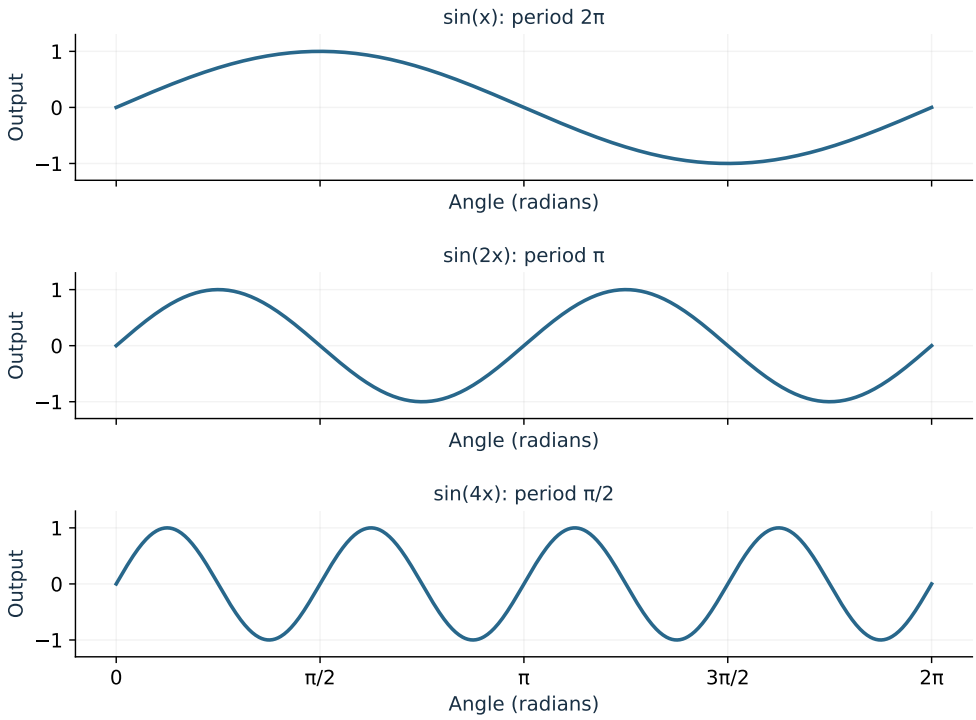


Figure 8: On the same horizontal axis, larger frequency coefficients fit more cycles into a fixed interval.

Frequency f counts cycles. Angular frequency $|\omega|$ counts radians during the same amount of time.

Worked example: revolutions per minute. A mechanism rotates 90 times each minute. That is $90/60 = 1.5$ cycles per second. Its period is $1/1.5 = 2/3$ s, and its angular speed is $2\pi(1.5) = 3\pi$ rad/s.

The sign of B reverses the direction of the inside angle. The period uses $|B|$. For ordinary sine models, a negative B can be absorbed into the sign of A.

3.4 Factor the inside expression to read a shift

A convenient form is

$$y = A\sin(B(x - C)) + D.$$

The graph has amplitude $|A|$, midline D, period $2\pi/|B|$, and a horizontal shift C. The expression $x - C$ becomes zero at $x = C$, so the familiar start of the sine cycle moves there.

The form of the parentheses matters. In $\sin(2(x - 1))$, the shift is 1. In $\sin(2x - 1)$, factor first:

$$\sin(2x - 1) = \sin\left(2\left(x - \tfrac{1}{2}\right)\right).$$

Its shift is $1/2$.

Worked example: read a transformed equation.

$$y = 2\sin(3x - \pi) + 1 = 2\sin\left(3\left(x - \frac{\pi}{3}\right)\right) + 1.$$

Its amplitude is 2, midline 1, period $2\pi/3$, and shift $\pi/3$ to the right. Its range is $[-1, 3]$.

The same ideas work for cosine. Remember its different starting value: a positive cosine coefficient begins a cycle at a maximum, while a positive sine coefficient begins at an upward midline crossing.

Try it. Rewrite $3\cos(5x - \pi) + 2$ in shifted form and identify its amplitude, period, shift, midline, and range.

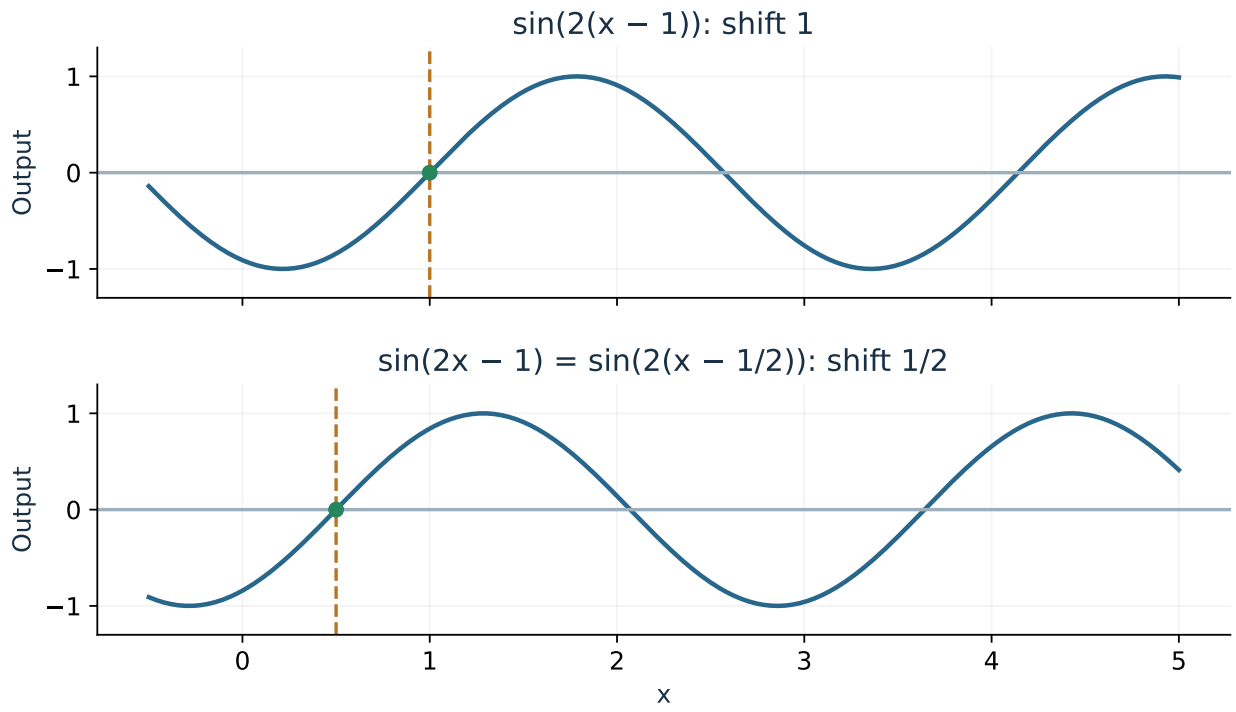


Figure 9: Factoring the entire inside expression identifies where the sine cycle begins. Equal periods can have different horizontal shifts.

3.5 Plot five landmarks before drawing the wave

Divide one period into four equal intervals. For $A\sin(B(x-C))+D$ with $B>0$, one cycle has the points

x	C	$C+T/4$	$C+T/2$	$C+3T/4$	$C+T$
y	D	$D+A$	D	$D-A$	D

For cosine, the outputs are $D+A, D, D-A, D, D+A$. Signed A is useful here: a negative A automatically reverses the high and low values.

Worked example: a reflected cosine. For $y=-2\cos(\pi(x-1))+3$, the period is 2. Beginning at $x=1$, the landmarks are

$$(1,1),\quad (3/2,3),\quad (2,5),\quad (5/2,3),\quad (3,1).$$

The cycle begins at a minimum because the cosine coefficient is negative. Draw a smooth wave through those points.

The Sinusoid Explorer marks these landmarks and lists their coordinates. Keep the viewing axes fixed while comparing shifts. Use its cycle view when you want to inspect a single period closely.

3.6 Build a model from a wheel

A wheel has radius 9 m, center 11 m above ground, and period 30 s. A passenger starts at the rightmost point and moves counterclockwise.

The center height gives the midline, the radius gives the amplitude, and one turn in 30 seconds gives $\omega=2\pi/30$. The passenger begins at the midline and initially rises, which fits a positive sine:

$$h(t)=11+9\sin\left(\frac{\pi t}{15}\right).$$

At $t = 0$, the height is 11 m. After a quarter-period, $t = 7.5$ s, the height is $11 + 9 = 20$ m. After three quarters, it is $11 - 9 = 2$ m.

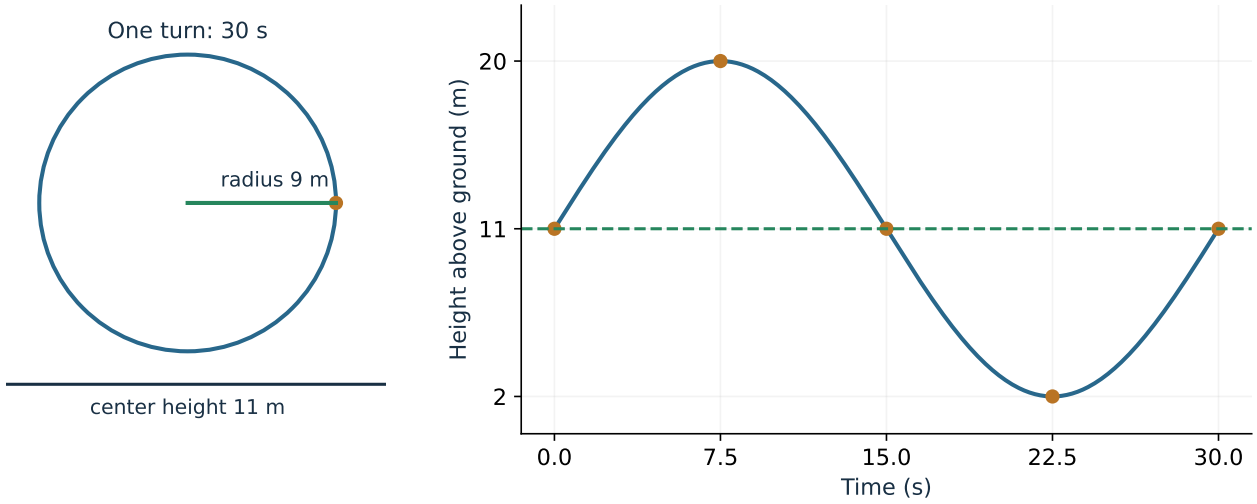


Figure 10: One revolution of the wheel produces one height cycle. The starting position determines whether a sine or cosine form is convenient.

If the passenger starts at the lowest point instead, a convenient model is $h(t) = 11 - 9\cos(\pi t/15)$. It begins at 2 m and reaches 20 m after half a period. The geometry determines the starting phase.

3.7 Build a model from measured data

Suppose a buoy has maximum height 5 m, minimum 1 m, and successive maxima 8 seconds apart. Taking $t = 0$ at a maximum gives

$$h(t) = 3 + 2\cos\left(\frac{\pi t}{4}\right).$$

The amplitude is 2 m, midline 3 m, and period 8 s. If a maximum occurs at $t = 2$ instead, replace t with $t - 2$.

To construct a model, identify the extremes, the repeat time, and a recognizable starting event. Choose sine or cosine to make that event easy to express. Then check the model at the starting time and at its expected extremes.

Try it. A tide ranges from 0.8 m to 3.6 m and repeats every 12 hours. Write a cosine model whose maximum occurs at $t = 2$ hours.

3.8 Extend the graph ideas to tangent and reciprocals

Tangent has base period π, so $y = A\tan(B(x - C)) + D$, with $A, B \neq 0$, has period $\pi/|B|$. Its asymptotes occur when $B(x - C) = \pi/2 + k\pi$. Tangent is unbounded, so it has no amplitude.

For secant and cosecant, begin with their cosine or sine denominator. The denominator's zeros locate the breaks. For example, $\sec x$ has asymptotes at $\pi/2 + k\pi$; its branches touch 1 and -1 at the extrema of cosine.

Finally, when $A = 0$, the sine or cosine model is constant. Every positive input shift preserves it, so it has no least positive period. A formula for the usual fundamental period applies to a nonconstant wave.

3.9 Review exercises

1. Find amplitude, midline, period, and range for $y = -4\sin(2x) + 1$.
2. Rewrite $3\cos(5x - \pi) + 2$ in shifted form.
3. Give five landmarks for $y = 2\sin(x - \pi/2)$.
4. Explain why $\sin(2x)$ completes two cycles on $[0, 2\pi]$.
5. Model a radius-6 wheel with center height 8 m and period 24 s, starting at the rightmost point.
6. Find the asymptotes of $\tan(2x)$ on $[0, 2\pi)$.
7. Sketch one period of $2\sec x$, marking the excluded inputs.
8. Explain how extremes determine amplitude and midline.
9. Give a sinusoidal model with maximum 8, minimum -2, period 10, and a maximum at $x = 3$.
10. Explain the period issue when $A = 0$.
11. An oscillation has frequency 2.5 cycles/s. Find period and angular frequency.
12. Give the domain of $\tan(3x)$ and two consecutive asymptotes.
13. Find amplitude and midline from maximum 12.4 and minimum 7.6.

3.10 Check your understanding

The first Try it has amplitude 6 and midline -1; one model is $6\sin x - 1$.

The shifted cosine has amplitude 3, period $2\pi/5$, shift $\pi/5$ right, midline 2, and range $[-1, 5]$.

For the tide, $h(t) = 2.2 + 1.4\cos\big((\pi/6)(t - 2)\big)$. The units are hours for t and meters for height; the model gives $h(2) = 3.6$.

4 Inverse Trigonometry and Identities

A ladder reaches 12 feet up a wall with its foot 5 feet from the wall. The ratio 12/5 tells us the tangent of the ladder's ground angle. How do we recover the angle itself?

An inverse trig function answers that question by choosing an angle with the requested ratio. Because trig functions repeat values, we must understand which angle it chooses. We will then use the geometry of coordinates and rotations to explain identities and calculate new exact values.

4.1 An inverse returns an angle

For the ladder, $\tan\theta = 12/5$, so

$$\theta = \arctan(12/5) \approx 67.4^\circ.$$

The acute angle supplied by the physical triangle fits the calculator's answer.

The notation $\tan^{-1}$ on a calculator means inverse tangent, also called arctangent. The inverse sine and cosine functions are written arcsin and arccos.

An inverse function reverses the input-output relationship. For example, $\sin 30^\circ = 1/2$. Reversing that relationship gives $\arcsin(1/2) = 30^\circ$. A reciprocal performs a different operation: $1/\sin 30^\circ = 2$.

Expression	What it asks	Result
$\arcsin(1/2)$	Which chosen angle has sine 1/2?	$\pi/6$ radians
$1/(1/2)$	What is the reciprocal of 1/2?	2
$\csc(\pi/6)$	What is the reciprocal of this angle's sine?	2

Always check degree or radian mode when a calculator reports an angle. The same geometric angle can be displayed as 30° or approximately 0.524 radians.

4.2 Restrict a repeating function before reversing it

Sine takes the value 1/2 at both $\pi/6$ and $5\pi/6$, and at infinitely many angles obtained by full turns. A function must return one output for each input, so inverse sine needs a convention that chooses one.

We restrict sine to $[-\pi/2, \pi/2]$. On that interval it rises steadily from -1 to 1 and takes each value exactly once. Reversing this restricted function defines arcsine.

The **principal ranges** specify the choices made by all three inverse functions:

Inverse function	Allowed inputs	Angle outputs
$\arcsin x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos x$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	All real numbers	$(-\pi/2, \pi/2)$

The tangent endpoints are excluded because tangent itself is undefined there. Its restricted branch still reaches every real output, so arctangent accepts every real input.

Worked example: choose the angle in the range. We have

$$\arcsin(-\sqrt{2}/2) = -\pi/4, \qquad \arccos(-1/2) = 2\pi/3, \qquad \arctan(-1) = -\pi/4.$$

Each output lies in its function's principal range and produces the requested trig value. For example, $5\pi/4$ also has tangent 1, but $\arctan(1) = \pi/4$.

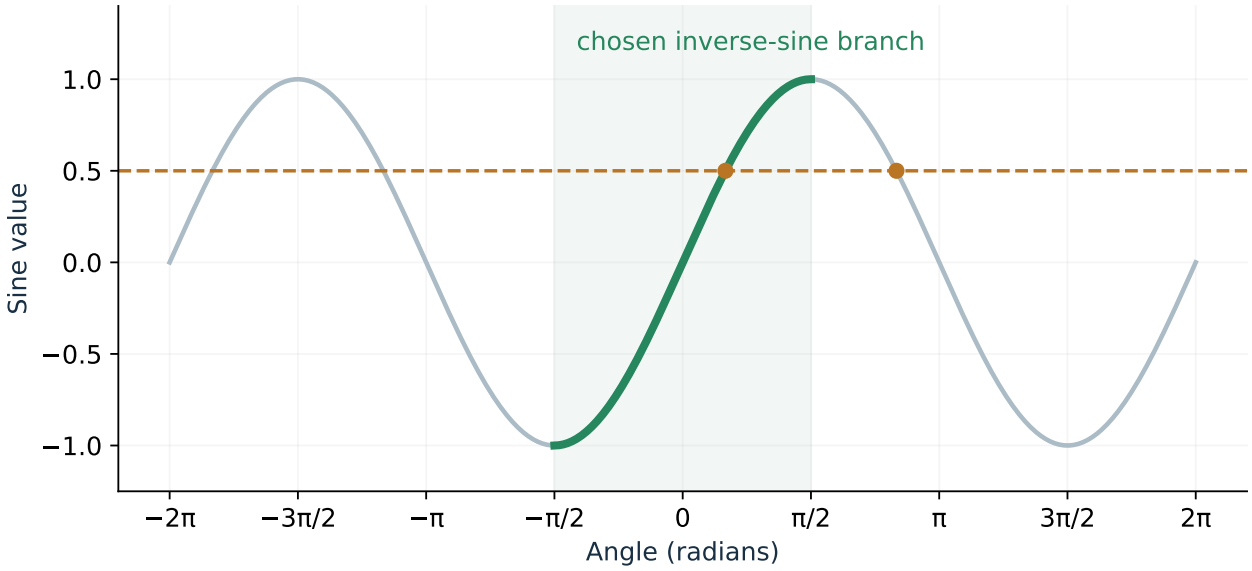


Figure 11: The green sine branch contains exactly one point at each height between minus one and one. The full sine curve repeats those heights.

The Inverse Trigonometry Explorer compares an original angle with the principal angle. Watch which information is retained when you keep only one trig value.

4.3 Reflect the restricted graph to see its inverse

If (a, b) lies on a function graph, the reversed relationship contains (b, a). Swapping coordinates reflects the graph across $y = x$.

For sine, use the restricted branch. The point $(\pi/6, 1/2)$ becomes $(1/2, \pi/6)$ on arcsine. The old range becomes the new domain, and the old domain becomes the new range.

Use the Inverse Graphs Explorer to compare all three inverse graphs. Arcsine and arccosine have bounded input intervals. Arctangent accepts all real inputs, while its output approaches the excluded angle endpoints.

Try it. Choose the cosine points at $0, \pi/2, \pi$. Swap their coordinates to obtain three points on arccosine.

4.4 Check the order in a composition

The expression $\sin(\arcsin u)$ starts with an allowed ratio u, turns it into the chosen angle, then takes that angle's sine. It returns u for every $u \in [-1, 1]$.

The expression $\arcsin(\sin \theta)$ starts with an angle. It first keeps only the sine value, then selects an angle in the principal range. It returns the original angle only when that angle already belongs to $[-\pi/2, \pi/2]$.

Worked example: two angles share one sine.

$$\arcsin(\sin 150^\circ) = \arcsin(1/2) = 30^\circ.$$

The original angle is outside the arcsine range. In contrast, $\arccos(\cos 150^\circ) = 150^\circ$, since 150° is in arccosine's range.

This distinction will matter when we solve equations in Module 5. An inverse trig function supplies a principal angle; an equation may require every angle producing the same value.

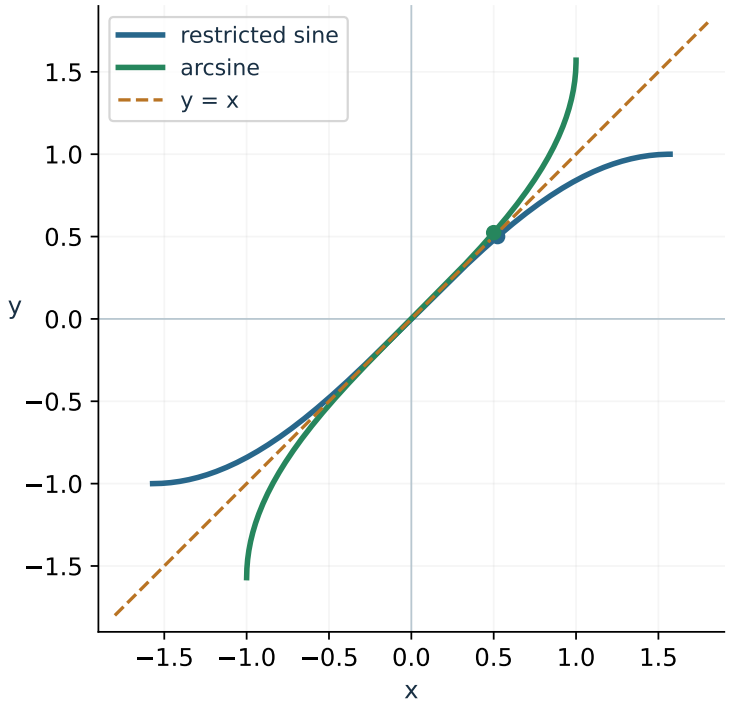


Figure 12: Swapping coordinates reflects the restricted sine curve across the diagonal to produce arcsine.

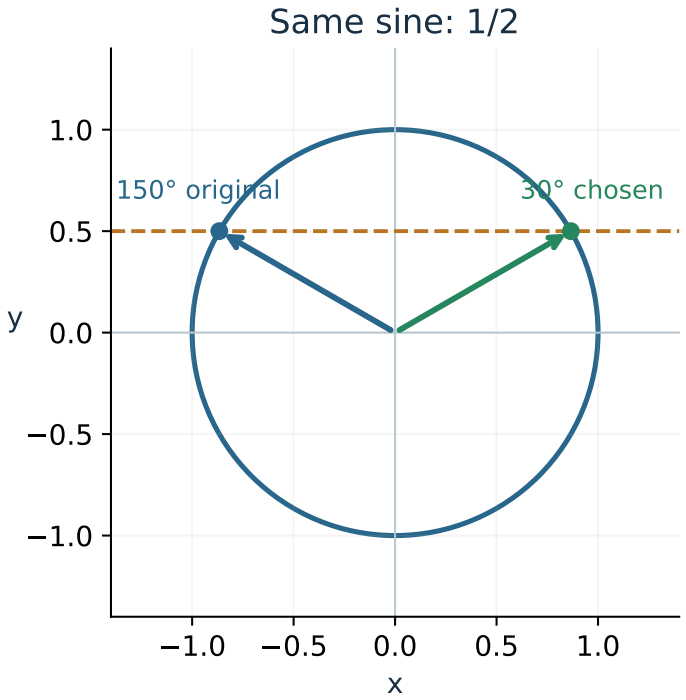


Figure 13: The 30-degree and 150-degree rays have the same height. Arcsine chooses the ray in its prescribed half-circle.

4.5 Use a triangle to evaluate mixed compositions

Consider $\sin(\arctan(3/4))$. Calling the inside angle θ makes the steps easier to see:

$$\theta = \arctan(3/4) \quad \Longrightarrow \quad \tan\theta = 3/4.$$

Since the ratio is positive and arctangent returns angles between $-\pi/2$ and $\pi/2$, this angle is acute. Draw a right triangle with opposite side 3 and adjacent side 4. Its hypotenuse is 5, giving $\sin\theta = 3/5$.

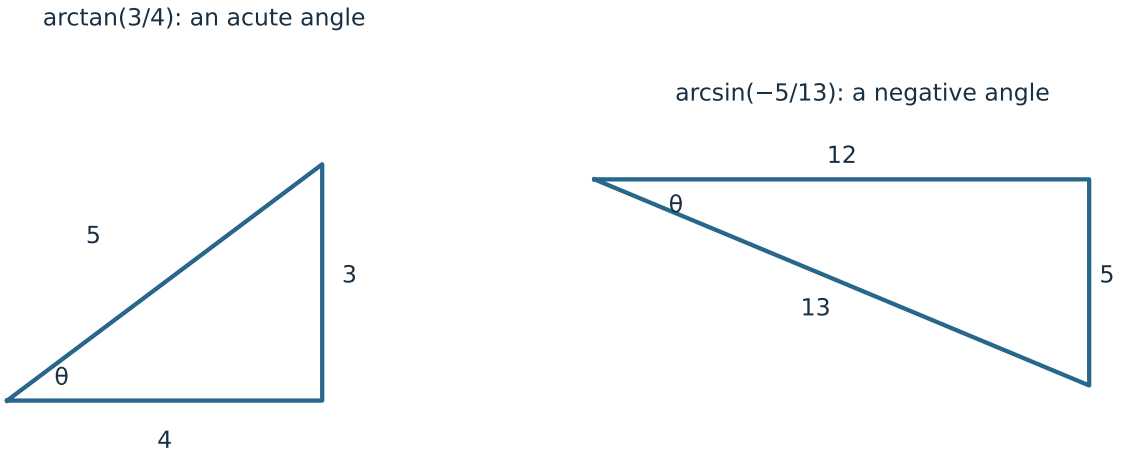


Figure 14: A ratio determines a reference triangle; the inverse function's range determines the coordinate signs.

Worked example: use the range to choose a sign. To find $\cos(\arcsin(-5/13))$, set $\theta = \arcsin(-5/13)$. Its sine is negative, and its principal angle lies between $-\pi/2$ and 0. Cosine is positive there:

$$\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-\frac{25}{169}} = \frac{12}{13}.$$

The square-root magnitude alone would leave a sign choice. The principal range resolves it.

Try it. Find $\sin(\arccos(-3/5))$ exactly. Identify the quadrant of the chosen angle before deciding the sign.

4.6 Derive identities from a familiar equation

An identity states that two expressions agree at every input where both are defined. A trig equation such as $\sin x = 1/2$ instead asks which inputs make the statement true.

The unit-circle equation gives our basic identity:

$$\sin^2 x + \cos^2 x = 1.$$

Divide every term by $\cos^2 x$, requiring $\cos x \neq 0$:

$$\frac{\sin^2 x}{\cos^2 x} + 1 = \frac{1}{\cos^2 x} \quad \Longrightarrow \quad 1 + \tan^2 x = \sec^2 x.$$

Dividing by $\sin^2 x$ similarly gives $1 + \cot^2 x = \csc^2 x$, where $\sin x \neq 0$.

Worked example: simplify while keeping the domain. For $\sin x \neq 0$,

$$\frac{1-\cos^2 x}{\sin x} = \frac{\sin^2 x}{\sin x} = \sin x.$$

The final expression exists at more inputs than the original quotient. Our identity retains the original restriction $x \neq k\pi$.

To prove an identity, start with one expression and use established algebraic or trigonometric relationships until you obtain the other. Record any denominator restrictions. Numerical agreement at a few angles helps check your work, but a proof must cover the whole stated domain.

4.7 Read symmetry identities from the circle

Reflecting a point across the horizontal axis preserves its horizontal coordinate and reverses its vertical coordinate. Therefore

$$\cos(-x) = \cos x, \qquad \sin(-x) = -\sin x.$$

Cosine is even and sine is odd. Their quotient gives $\tan(-x) = -\tan x$ wherever tangent is defined.

A half-turn negates both coordinates:

$$\cos(x+\pi) = -\cos x, \qquad \sin(x+\pi) = -\sin x.$$

For example, $\sin(-7\pi/6) = -\sin(7\pi/6) = 1/2$, while $\cos(-7\pi/6) = \cos(7\pi/6) = -\sqrt{3}/2$.

4.8 Combine two rotations to derive addition formulas

An initial turn of β gives the point $(\cos\beta, \sin\beta)$. Now rotate that entire configuration by another angle α.

The original horizontal unit direction turns to $(\cos\alpha, \sin\alpha)$. The original vertical unit direction, a quarter-turn farther around, turns to $(-\sin\alpha, \cos\alpha)$.

The point's horizontal contribution has length $\cos\beta$ along the first direction. Its vertical contribution has length $\sin\beta$ along the second. Add their coordinate contributions:

$$x = \cos\beta\cos\alpha - \sin\beta\sin\alpha,$$
$$y = \cos\beta\sin\alpha + \sin\beta\cos\alpha.$$

The final point also has angle $\alpha+\beta$. Its coordinates are cosine and sine of that total angle.

Consequently,

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Replacing β by $-\beta$ and using symmetry gives the difference formulas:

$$\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta,$$
$$\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta.$$

Worked example: an exact value at a new angle. Write $75^\circ = 45^\circ + 30^\circ$. Then

$$\sin 75^\circ = \frac{\sqrt{2}}{2}\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\frac{1}{2} = \frac{\sqrt{6}+\sqrt{2}}{4}.$$

Similarly, $\cos 75^\circ = (\sqrt{6}-\sqrt{2})/4$. Both are positive, as expected in Quadrant I.

The formulas explain why $\sin(a+b)$ generally differs from $\sin a + \sin b$. Each original coordinate contributes to both final coordinates after rotation.

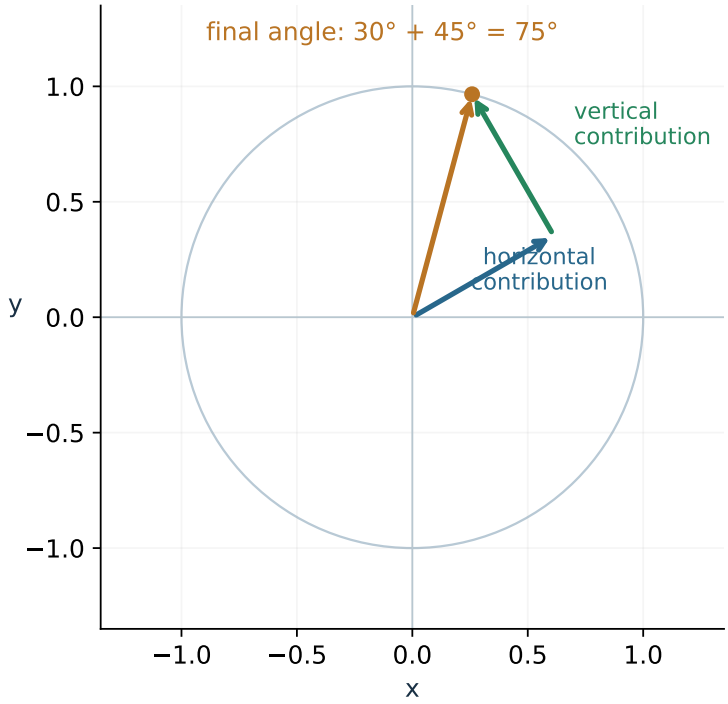


Figure 15: The two rotated coordinate contributions add to the point at the sum of the angles.

4.9 Double angles, squared functions, and half angles

Set both angles equal to x:

$$\sin 2x = 2\sin x \cos x, \qquad \cos 2x = \cos^2 x - \sin^2 x.$$

Using the Pythagorean identity gives two more cosine forms:

$$\cos 2x = 1 - 2\sin^2 x = 2\cos^2 x - 1.$$

Choose the form containing the quantities you already know.

Rearranging produces the power-reduction identities

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \qquad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

Replacing x by $\theta/2$ gives the squared half-angle values. Taking square roots requires the sign from the half-angle's quadrant. For instance, $15°$ is acute, so

$$\cos 15° = \sqrt{\frac{1 + \cos 30°}{2}} = \frac{\sqrt{2 + \sqrt{3}}}{2}.$$

This equals $(\sqrt{6} + \sqrt{2})/4$, the form obtained from the cosine difference identity.

Try it. Use angle addition to find $\sin 105°$. Then find $\cos 2x$ using $\sin x$ alone.

4.10 Review exercises

1. Evaluate $\arcsin(-\sqrt{2}/2)$ and $\arctan(\sqrt{3})$ exactly.
2. Explain the output of $\arcsin(\sin(3\pi/4))$.
3. Find $\cos(\arcsin(5/13))$ using a triangle.
4. Find $\sin(\arctan(-2))$, including its sign.
5. Prove $(\sec^2 x - 1)/\tan x = \tan x$ on its domain.
6. Prove $(\sin x + \cos x)^2 = 1 + \sin 2x$.
7. Calculate $\cos 75^\circ$ exactly.
8. Express $\cos 2x$ using sine alone.
9. State and explain the domain and range of arccosine.
10. Evaluate $\arccos(\cos(5\pi/3))$.
11. Find $\tan(\arccos(4/5))$ exactly.
12. Use circle symmetry to find $\sin(-\pi/3)$.
13. Derive the cosine difference identity from the sum identity.
14. Prove $\cos^2 x = (1 + \cos 2x)/2$.
15. Find $\sin 15^\circ$ exactly and state its sign.
16. Explain the output of $\arctan(\tan(3\pi/4))$.

4.11 Check your understanding

The reflected cosine points are $(1, 0), (0, \pi/2), (-1, \pi)$. For the mixed composition, arccosine selects a Quadrant II angle, so $\sin(\arccos(-3/5)) = 4/5$.

The addition calculation gives $\sin 105^\circ = (\sqrt{6} + \sqrt{2})/4$, positive in Quadrant II. The requested double-angle form is $\cos 2x = 1 - 2\sin^2 x$.

5 Trigonometric Equations and Applications

An inverse trig button gives an angle. An equation can ask for several angles, or for an entire repeating family. The difference comes from the geometry we have been studying: a rotating point can return to the same height or horizontal position more than once.

We will use that picture to solve equations, then use the same trigonometric relationships to find distances in triangles. Keep a sketch beside your algebra. It helps you decide how many answers to expect and whether your answers fit the situation.

5.1 Find the angles that give a chosen height

Consider $\sin x = \frac{1}{2}$. Sine is the vertical coordinate on the unit circle, so we want every point on the circle whose height is $\frac{1}{2}$. A horizontal line at that height meets the circle twice. The angles are $\pi/6$ and $5\pi/6$.

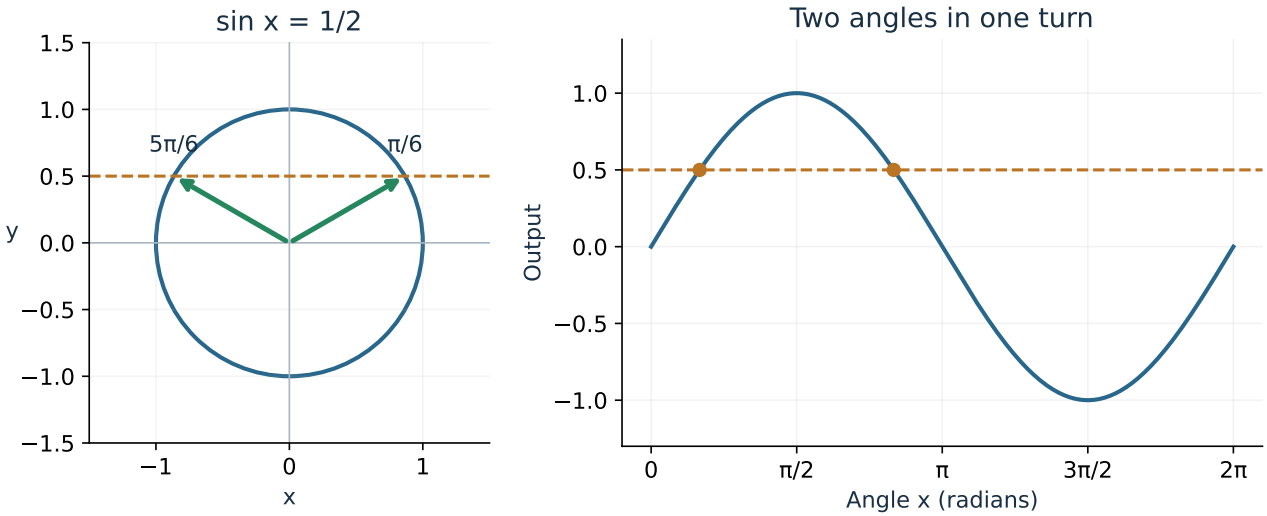


Figure 16: A horizontal line finds two points on the circle and two intersections on one sine cycle.

On the interval $0 \le x < 2\pi$, those are the two answers. The left endpoint is included; the right endpoint is excluded. That choice represents one complete turn without counting its starting position twice.

If the question asks for **all real solutions**, each angle may be followed by any whole number of turns:

$$x = \frac{\pi}{6} + 2\pi k \quad \text{or} \quad x = \frac{5\pi}{6} + 2\pi k, \qquad k \in \mathbb{Z}.$$

The integer k includes turns in either direction. For example, $k = -1$ gives negative angles that reach the same two points.

Worked example: a negative horizontal coordinate. Solve $\cos x = -\sqrt{2}/2$ on $[0, 2\pi)$. The reference angle is $\pi/4$. Cosine is negative on the left half of the circle, in Quadrants II and III. Therefore

$$x = \pi - \frac{\pi}{4} = \frac{3\pi}{4}, \qquad x = \pi + \frac{\pi}{4} = \frac{5\pi}{4}.$$

Both answers give the requested horizontal coordinate.

For tangent, a half-turn restores the same slope. Thus $\tan x = -1$ has the family $x = -\pi/4 + k\pi$. On $[0, 2\pi)$, the two members of that family are $3\pi/4$ and $7\pi/4$.

Try it. Solve $\sin x = -\sqrt{3}/2$ on $[0, 2\pi)$. Mark the two points before writing their angles.

5.2 Solve for the whole inside angle

In $\sin(2x) = \sqrt{3}/2$, the input to sine is $2x$. Call that input u. As x runs through $[0, 2\pi)$, the angle $u = 2x$ runs through $[0, 4\pi)$. We must search two turns.

On the first turn, $u = \pi/3$ or $2\pi/3$. Add 2π to get the second-turn answers $7\pi/3$ and $8\pi/3$. Now divide **each** by 2:

$$x = \frac{\pi}{6}, \quad \frac{\pi}{3}, \quad \frac{7\pi}{6}, \quad \frac{4\pi}{3}.$$

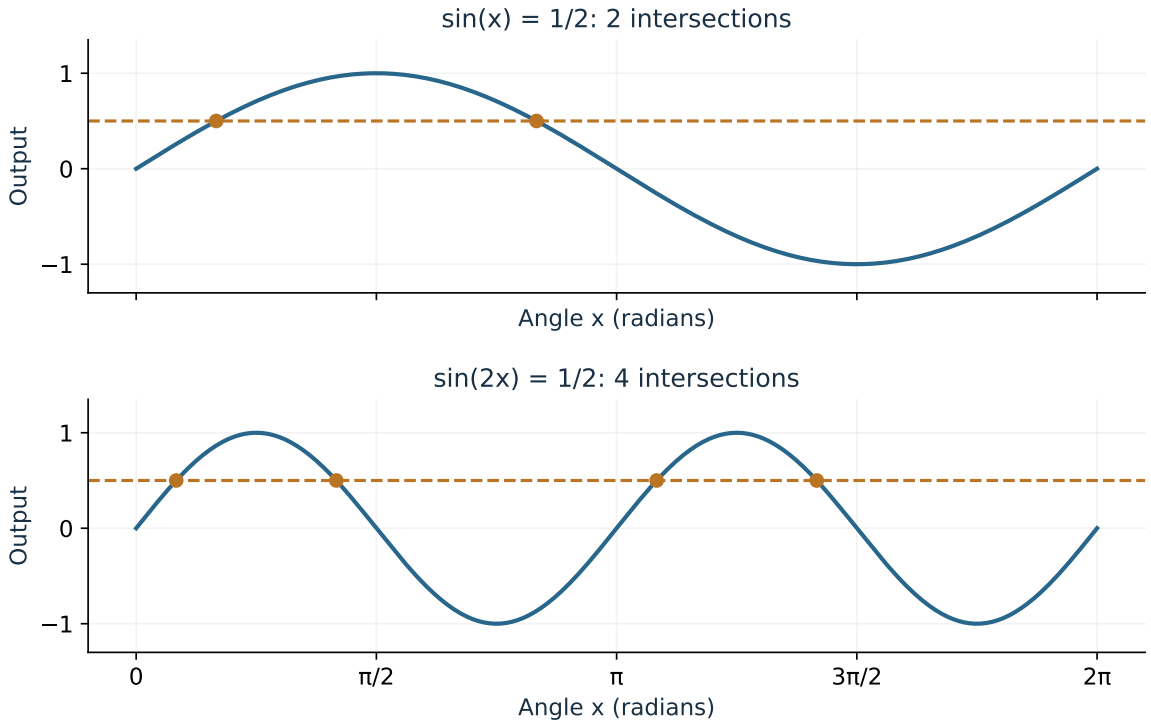


Figure 17: Doubling the inside angle fits two sine cycles into the same interval and doubles the intersections with height one half.

The graph provides a useful count: $\sin(2x)$ completes two cycles on this interval, and the chosen positive height occurs twice per cycle. We expect four solutions. At a maximum or minimum there is just one intersection per cycle; a target outside $[-1, 1]$ gives none.

Worked example: a shifted inside angle. Solve $\sin(x - \pi/4) = 1/2$ on $[0, 2\pi)$. Set $u = x - \pi/4$; its interval is $[-\pi/4, 7\pi/4)$. In that interval, $u = \pi/6$ or $5\pi/6$. Adding $\pi/4$ gives

$$x = \frac{5\pi}{12} \quad \text{or} \quad \frac{13\pi}{12}.$$

Substituting either answer restores one of the two familiar sine angles.

Use the Equation Explorer to compare frequencies 1, 2, and 3. Predict the number of intersections before displaying them.

5.3 Let algebra organize the choices

Start an equation such as $2\cos x - 1 = 0$ by isolating its trig expression. We obtain $\cos x = 1/2$, then use the circle to find $x = \pi/3, 5\pi/3$ on one turn.

Equations that are quadratic in a trig expression can be factored just as quadratics in an ordinary variable:

$$2\sin^2 x - \sin x = 0 \quad \Longrightarrow \quad \sin x(2\sin x - 1) = 0.$$

The zero-product rule gives two branches:

$$\sin x = 0 \quad \text{or} \quad \sin x = \frac{1}{2}.$$

The first branch contributes $0, \pi$; the second contributes $\pi/6, 5\pi/6$. There are four distinct solutions on $[0, 2\pi)$. Dividing the original equation by $\sin x$ would remove the first branch. Factoring keeps it visible.

An identity can also reveal a familiar equation. For $2\cos^2 x - 1 = 0$, we get $\cos x = \pm\sqrt{2}/2$. Each sign contributes two angles. When you use a quotient, reciprocal, or division, keep track of inputs excluded from the original expression. Check your final angles in that original expression.

Try it. Solve $2\cos^2 x - \cos x = 0$ on $[0, 2\pi)$. First write the two equations produced by factoring.

5.4 Label a triangle before choosing a law

For any triangle, use capital letters A, B, C for angles and the corresponding lowercase letters a, b, c for the sides **opposite** them. Angle A and side a are separated across the triangle. Angles add to $180°$, or π radians.

Right-triangle ratios use an opposite side, adjacent side, and hypotenuse. For a triangle with no right angle, we need relationships that work without a hypotenuse.

In the pictured triangle, the altitude has height $h = b\sin A = a\sin B$. Equating these expressions and rearranging gives $a/\sin A = b/\sin B$. The same reasoning with another altitude gives the **Law of Sines**:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

It is especially useful when an angle and its opposite side are both known. That complete pair supplies a ratio to compare with another pair.

Worked example: two angles and a side. Suppose $A = 40°$, $B = 65°$, and $a = 9$ cm. First find $C = 180° - 40° - 65° = 75°$. Pair a with A, and b with B:

$$\frac{b}{\sin 65°} = \frac{9}{\sin 40°}, \qquad b = \frac{9\sin 65°}{\sin 40°} \approx 12.69 \text{ cm}.$$

The larger angle $65°$ faces the longer side, which agrees with the result. Use degree mode for these measurements.

5.5 Why two triangles can fit the same measurements

Suppose $A = 30°$, $a = 5$, and $b = 8$. The known angle is opposite the side of length 5. The Law of Sines gives

$$\sin B = \frac{b\sin A}{a} = \frac{8\sin 30°}{5} = 0.8.$$

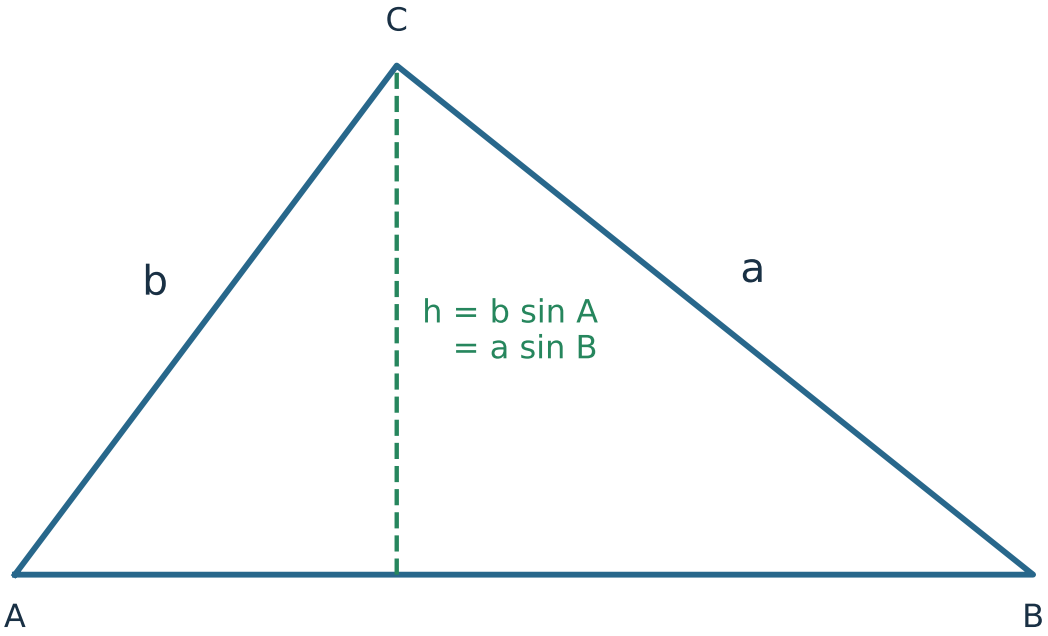


Figure 18: An altitude belongs to two right triangles. Expressing its height in two ways produces the Law of Sines.

The calculator gives $B_1 = \arcsin(0.8) \approx 53.13°$. Sine also has that value at its supplement, $B_2 = 180° - B_1 \approx 126.87°$. Check each candidate against the angle sum:

$$C_1 \approx 96.87°, \qquad C_2 \approx 23.13°.$$

Both remaining angles are positive, so both triangles work.

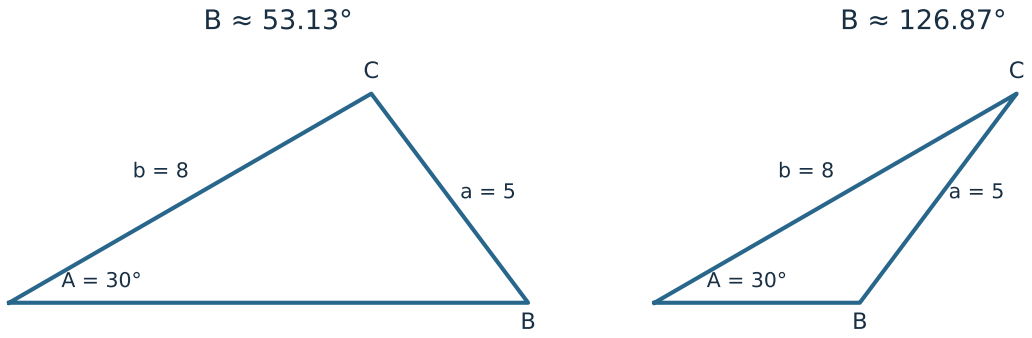


Figure 19: The same angle and two side lengths can form a shorter triangle or a longer triangle.

This is the **ambiguous SSA case**: two sides and a nonincluded angle are given. Follow three checks. If the calculated sine exceeds 1, no triangle exists. Otherwise find the principal angle and its supplement. Keep only distinct candidates for which $A + B < 180°$. Depending on the measurements, zero, one, or two triangles survive.

For example, changing a to 3 gives $\sin B = 4/3$, so there is no triangle. Changing a to 8 gives candidates 30° and 150°; the latter leaves $C = 0^\circ$, so only the first is a triangle.

5.6 Use the included angle with the Law of Cosines

If two sides and the angle **between** them are known, there is no complete opposite side-angle pair. The **Law of Cosines** supplies the missing side:

$$c^2 = a^2 + b^2 - 2ab\cos C.$$

To see where it comes from, place angle C at the origin. One endpoint is $(b, 0)$, and the other is $(a\cos C, a\sin C)$. Their squared distance is

$$\begin{aligned} c^2 &= (a\cos C - b)^2 + (a\sin C)^2 \\ &= a^2(\cos^2 C + \sin^2 C) + b^2 - 2ab\cos C. \end{aligned}$$

The Pythagorean identity reduces the first term to a^2.

Worked example: a connector between two roads. Two roads extend 7 km and 10 km from a junction, separated by 60°. The connector between their endpoints has length

$$c = \sqrt{7^2 + 10^2 - 2(7)(10)\cos 60^\circ} = \sqrt{79} \approx 8.89 \text{ km}.$$

At 90°, the cosine term is zero and the formula becomes the Pythagorean theorem. For the same side lengths, a wider included angle produces a longer connector.

If all three sides are known, rearrange the law to recover an angle:

$$C = \arccos\left(\frac{a^2 + b^2 - c^2}{2ab}\right).$$

The geometric angle lies between 0° and 180°, which matches the range of arccosine.

5.7 Choose a method and interpret the result

For an equation, decide which interval to search, solve for the trig expression, and use periodicity to find every answer. For a triangle, draw and label the known measurements. A complete opposite pair suggests the Law of Sines; two sides with their included angle, or three sides, suggest the Law of Cosines.

A shoreline survey illustrates the choice. Two observers stand 200 m apart and measure interior sight angles of 47° and 68° to an island marker. The third angle is 65°. The known baseline is opposite that angle, so it supplies the complete pair for the Law of Sines. The geometry tells us which measurements belong together before we calculate.

5.8 Review exercises

1. Solve $\sin x = -\sqrt{3}/2$ on $[0, 2\pi)$, then write all real solutions.
2. Solve $2\sin(3x) = 1$ on $[0, 2\pi)$. Explain the number of answers.
3. Solve $2\cos^2 x - \cos x = 0$ on $[0, 2\pi)$.
4. Solve $\tan x = -1$ for all real x.
5. Solve $\cos(2x) = 1/2$ on $[0, 2\pi)$.
6. A triangle has $A = 35°$, $B = 80°$, $a = 12$. Find C and b.
7. For $A = 30°$, $a = 5$, $b = 8$, find both possible values of C.
8. Find the third side for sides 6 and 9 with included angle $120°$.
9. Explain how the Law of Cosines includes the Pythagorean theorem.
10. Solve $\sin(x - \pi/6) = 1/2$ on $[0, 2\pi)$.

5.9 Check your understanding

The first Try it has answers $4\pi/3, 5\pi/3$. For the factoring Try it, $\cos x = 0$ or $\cos x = 1/2$, giving $\pi/2, 3\pi/2, \pi/3, 5\pi/3$. Each branch contributes its own angles. A sketch or a substitution checks the result.

6 Vectors and Geometric Applications

A person walks 6 km east, then 8 km north. The route is 14 km long, but the endpoint is only 10 km from the start. To describe where the person ends up, we need both a distance and a direction. A vector records those two pieces of information together.

This chapter develops vectors from familiar coordinates and triangle geometry. We will add movements, resolve forces into components, and measure how much one vector points along another.

6.1 Describe a displacement with components

The displacement $\mathbf{u} = \langle 6, 8\rangle$ means 6 units horizontally and 8 units vertically. Angle brackets distinguish a vector from a location $(6, 8)$. The same components can describe an arrow starting anywhere in the plane.

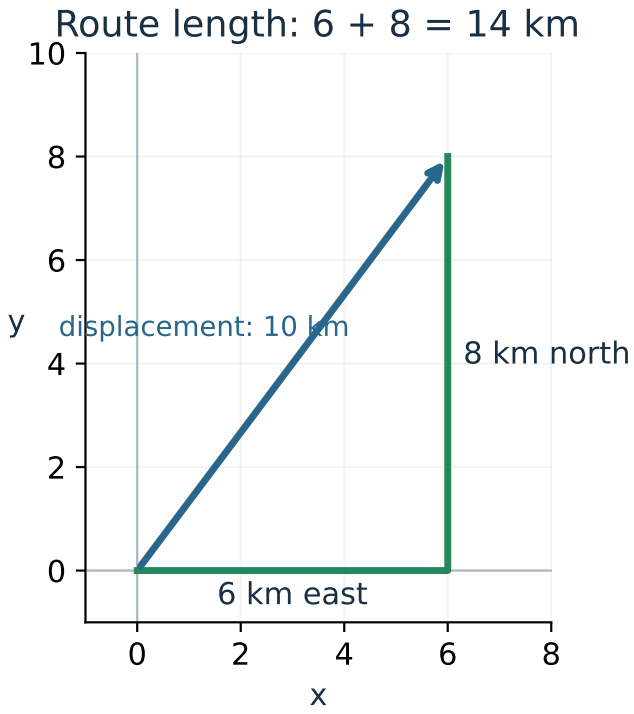


Figure 20: The walking route is 14 km; the displacement joins the start directly to the endpoint.

For a vector $\mathbf{u} = \langle u_x, u_y\rangle$, its **magnitude** is the length of its arrow:

$$\|\mathbf{u}\| = \sqrt{u_x^2 + u_y^2}.$$

For the walk, $\sqrt{6^2 + 8^2} = 10$ km. This is displacement magnitude. The distance walked depends on the route.

To find the vector from $P = (p_x, p_y)$ to $Q = (q_x, q_y)$, subtract start from finish:

$$\overrightarrow{PQ} = \langle q_x - p_x, q_y - p_y\rangle.$$

Starting at P, adding those two changes lands exactly at Q.

Worked example: from one location to another. For $P = (-2, 3)$, $Q = (5, -1)$,

$$\overrightarrow{PQ} = \langle 5 - (-2), -1 - 3\rangle = \langle 7, -4\rangle, \qquad \|\overrightarrow{PQ}\| = \sqrt{65}.$$

The endpoint lies 7 units right and 4 units down from the start. Reversing the journey gives $\overrightarrow{QP} = \langle -7, 4 \rangle$.

6.2 Add arrows by following one with another

If a movement $\mathbf{u}$ is followed by $\mathbf{v}$, the total horizontal change is $u_x + v_x$, and the total vertical change is $u_y + v_y$:

$$\mathbf{u} + \mathbf{v} = \langle u_x + v_x, u_y + v_y \rangle.$$

Draw the second arrow with its tail at the first arrow's head. The sum joins the original start to the final endpoint. Translating an arrow for this construction preserves its components.

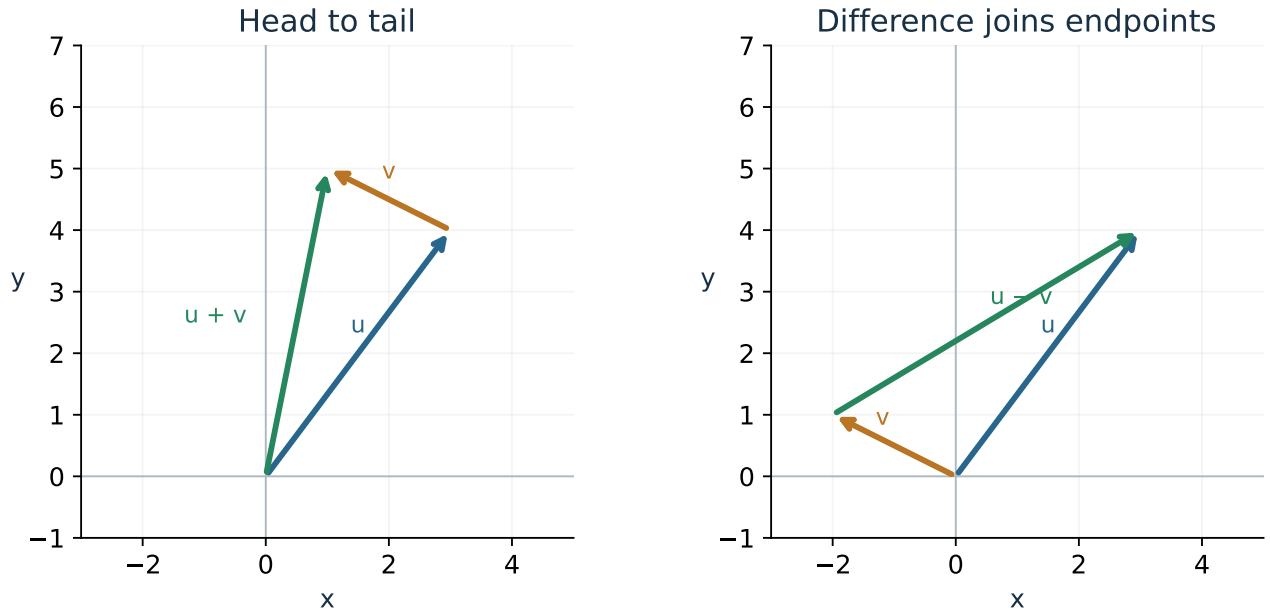


Figure 21: Head-to-tail addition and the parallelogram give the same sum. The difference joins the endpoints when both arrows start together.

For $\mathbf{u} = \langle 3, 4 \rangle$ and $\mathbf{v} = \langle -2, 1 \rangle$, the sum is $\langle 1, 5 \rangle$. The second movement goes left by 2 and up by 1. Adding it reduces the eastward component and increases the northward component.

Multiplying by a number scales every component:

$$c\mathbf{u} = \langle cu_x, cu_y \rangle.$$

For $c > 0$, the arrow keeps its direction; for $c < 0$, it reverses direction. Its length becomes $|c|\|\mathbf{u}\|$. The zero vector $\langle 0, 0 \rangle$ has magnitude zero and has no direction.

Subtraction means adding the opposite: $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$. If both vectors start at the origin, $\mathbf{u} - \mathbf{v}$ points from the endpoint of $\mathbf{v}$ to the endpoint of $\mathbf{u}$.

Try it. With $\mathbf{u} = \langle 2, 5 \rangle$, $\mathbf{v} = \langle -1, 3 \rangle$, calculate $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$. Check the direction of each component.

6.3 Convert between length-and-angle and components

A vector of magnitude $r \geq 0$ at angle θ, measured counterclockwise from the positive horizontal axis, has components

$$\mathbf{u} = \langle r \cos\theta, r \sin\theta \rangle.$$

This is the unit-circle point scaled by r. The magnitude returns to r, because $\cos^2\theta + \sin^2\theta = 1$.

Worked example: a force pointing up and left. A 20 N force acts at 150°:

$$\mathbf{F} = \langle 20\cos 150^\circ, 20\sin 150^\circ\rangle = \langle -10\sqrt{3}, 10\rangle \text{ N}.$$

The negative horizontal component describes the leftward part of the force; the positive vertical component describes its upward part.

For a vector with known components, find the magnitude first. To recover direction, inspect the quadrant before using inverse tangent. For $\langle -3, 4\rangle$, the reference angle is $\arctan(4/3) \approx 53.13^\circ$. The vector lies in Quadrant II, so its direction is $180^\circ - 53.13^\circ = 126.87^\circ$. The single quotient $4/(-3)$ does not by itself specify that quadrant.

A **unit vector** has magnitude 1. For a nonzero vector,

$$\widehat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|}.$$

Thus $\langle -3, 4\rangle/5 = \langle -3/5, 4/5\rangle$ keeps the direction and sets the length to 1.

6.4 The dot product measures alignment

The dot product combines two vectors into a number:

$$\mathbf{u}\cdot\mathbf{v} = u_x v_x + u_y v_y.$$

For nonzero vectors, this number also has a geometric form:

$$\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos\phi,$$

where $0 \le \phi \le \pi$ is the angle between their arrows.

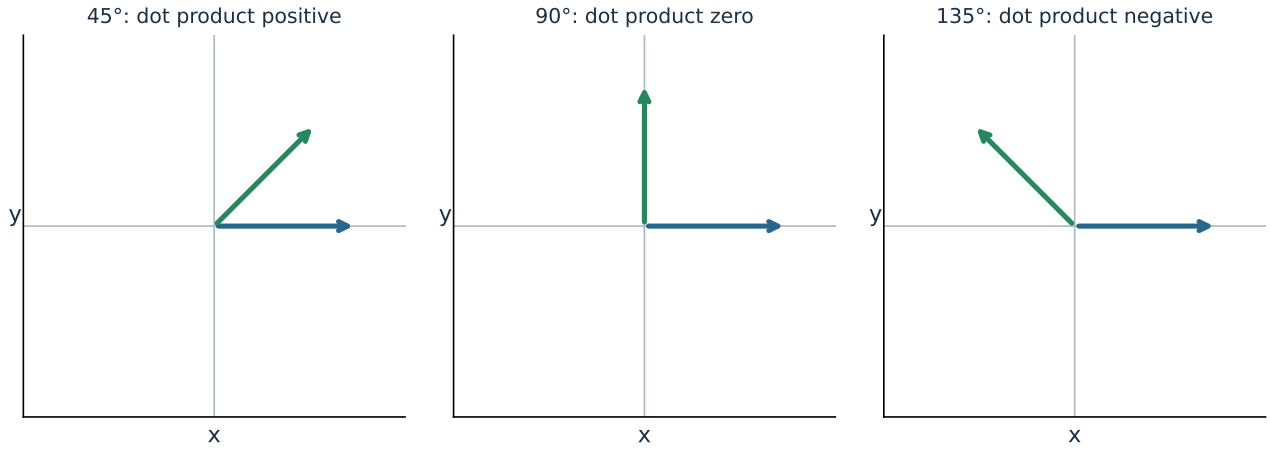


Figure 22: For vectors of fixed lengths, the dot product is positive at an acute angle, zero at a right angle, and negative at an obtuse angle.

The sign has meaning. Positive indicates an acute angle, zero indicates a right angle, and negative indicates an obtuse angle. A zero vector has dot product zero with every vector, but there is no angle to assign to it.

Worked example: perpendicular vectors. Let $\mathbf{u} = \langle 3, 4\rangle$, $\mathbf{v} = \langle 4, -3\rangle$. Their dot product is $12 - 12 = 0$. Both are nonzero, so the angle between them is 90°.

To calculate another angle, solve the geometric formula:

$$\phi = \arccos\left(\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}\right).$$

For $\langle 2,1\rangle$ and $\langle 1,2\rangle$, the dot product is 4 and each magnitude is $\sqrt{5}$. Thus $\phi = \arccos(4/5) \approx 36.87°$.

6.5 A projection is the part pointing along a chosen direction

Imagine a light casting an arrow's shadow onto a line through the origin. The **projection** records the part of $\mathbf{u}$ along the direction of a nonzero vector $\mathbf{v}$:

$$\operatorname{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|^2}\mathbf{v}.$$

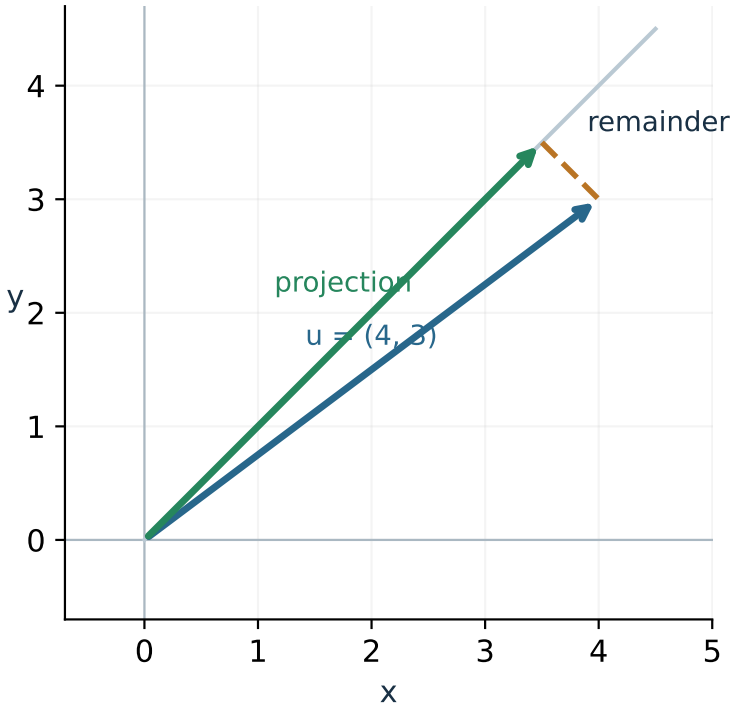


Figure 23: Projection separates a vector into a part parallel to the chosen direction and a perpendicular remainder.

Why this coefficient? First divide $\mathbf{v}$ by its length to get a unit direction $\hat{\mathbf{v}}$. The signed length of the shadow is $\mathbf{u}\cdot\hat{\mathbf{v}} = \|\mathbf{u}\|\cos\phi$. Multiply that length by $\hat{\mathbf{v}}$, obtaining the displayed vector formula.

Worked example: horizontal and slanted projections. Projecting $\langle 4,3\rangle$ onto $\langle 1,0\rangle$ gives $\langle 4,0\rangle$. Onto $\mathbf{v} = \langle 1,1\rangle$, the coefficient is $(4+3)/(1+1) = 7/2$, giving $\langle 7/2, 7/2\rangle$. The remainder is $\langle 1/2, -1/2\rangle$; its dot product with $\langle 1,1\rangle$ is zero.

6.6 Work uses the force along the displacement

For a constant force $\mathbf{F}$ acting through displacement $\mathbf{d}$, the work is

$$W = \mathbf{F}\cdot\mathbf{d} = \|\mathbf{F}\|\|\mathbf{d}\|\cos\phi.$$

Newtons times meters give joules. A perpendicular force contributes zero work to that displacement. A force pointing against the displacement contributes negative work.

A force $\langle 12, 5\rangle$ N acting through $\langle 3, 0\rangle$ m does $12(3) + 5(0) = 36$ J of work. Only the horizontal component contributes. If the force instead acted at 60° to the displacement, only half its magnitude would act along the movement.

Try it. A 50 N force acts at 60° to a 4 m displacement. Calculate its work and compare with the work from a parallel force of the same magnitude.

6.7 The triangle law returns in vector language

Place two vectors $\mathbf{u}, \mathbf{v}$ at a common starting point. Their difference joins their endpoints. Expanding its squared magnitude gives

$$\begin{aligned}\|\mathbf{u} - \mathbf{v}\|^2 &= (\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\mathbf{u} \cdot \mathbf{v} \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\| \cos\phi.\end{aligned}$$

This is the Law of Cosines. The sides are the two original vectors and their connecting difference. It explains why component calculations and triangle calculations agree.

Use the Vector Explorer to compare the picture with the component arithmetic. When you change one component, describe the movement of the endpoint before reading the new magnitude or dot product.

6.8 Review exercises

1. Find the vector from $(-2, 3)$ to $(5, -1)$, and its magnitude.
2. For $\mathbf{u} = \langle 2, 5\rangle$, $\mathbf{v} = \langle -1, 3\rangle$, compute $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$.
3. Find exact components of a displacement of magnitude 12 at 150°.
4. Find a unit vector in the direction $\langle -3, 4\rangle$.
5. Compute $\langle 2, -1\rangle \cdot \langle 5, 3\rangle$ and interpret its sign.
6. Determine whether $\langle 6, 2\rangle$ and $\langle -1, 3\rangle$ are perpendicular.
7. Find the projection of $\langle 4, 3\rangle$ onto $\langle 1, 0\rangle$.
8. Find the work of a 50 N force acting at 60° to a 4 m displacement.
9. Find the angle between $\langle 1, 0\rangle$ and $\langle 1, \sqrt{3}\rangle$.
10. Give two routes with the same displacement but different total distances.

6.9 Check your understanding

The first Try it gives $\langle 1, 8\rangle$ and $\langle 8, 9\rangle$. The work example gives $50(4)\cos 60^\circ = 100$ J, compared with 200 J for a parallel force. The factor $\cos 60^\circ = 1/2$ measures the contributing part of the force.

7 Polar Coordinates and Complex Numbers

An address such as $(3, 4)$ describes a point by moving horizontally and vertically. A distance of 5 at a direction of about $53°$ describes the same point by aiming and traveling. Polar coordinates use that second description.

Complex numbers will add another interpretation. They represent points in the plane, and their multiplication performs rotations and scaling. The trigonometry from earlier chapters explains why.

7.1 Locate a point by distance and direction

A polar pair (r, θ) gives a radius and an angle from the positive horizontal axis. For $r > 0$, start along the ray at angle θ and travel r units from the origin. The familiar circle coordinates give

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

These are the same component formulas used for vectors. Here they locate a point.

Worked example: a point in Quadrant II. For $(r, \theta) = (4, 5\pi/6)$,

$$x = 4(-\sqrt{3}/2) = -2\sqrt{3}, \qquad y = 4(1/2) = 2.$$

The negative x and positive y agree with the angle's quadrant. The point is 4 units from the origin.

To go in the other direction with $r \geq 0$, use $r = \sqrt{x^2 + y^2}$. Find the angle from both coordinate signs. For $(-3, -3)$, the distance is $3\sqrt{2}$, and the direction is $5\pi/4$ on $[0, 2\pi)$. Computing $\arctan(y/x) = \pi/4$ alone would place the point in the wrong quadrant.

Try it. Convert $(6, -\pi/3)$ to Cartesian coordinates. Give the same direction using an angle between 0 and 2π.

7.2 Several polar addresses can name one point

Adding a full turn changes an angle's description while leaving its direction fixed:

$$(r, \theta) \quad \text{and} \quad (r, \theta + 2\pi k)$$

represent the same point for every integer k.

A negative radius means travel in the direction opposite the chosen ray. Consequently,

$$(r, \theta) \quad \text{and} \quad (-r, \theta + \pi)$$

also represent the same point.

For example, $(2, \pi/3)$, $(2, 7\pi/3)$, and $(-2, 4\pi/3)$ all locate $(1, \sqrt{3})$. Check the last address directly: both cosine and sine at $4\pi/3$ are negative, and multiplying by -2 makes the coordinates positive.

At $r = 0$, all choices of angle locate the origin. This redundancy is part of the coordinate system. A unique polar address requires conventions about radius and angle.

7.3 Put complex numbers on the plane

A complex number $z = a + bi$, where $i^2 = -1$, corresponds to the point (a, b). The horizontal axis records its real part a, and the vertical axis its imaginary part b.

Its **modulus** is its distance from zero:

$$|z| = \sqrt{a^2 + b^2}.$$

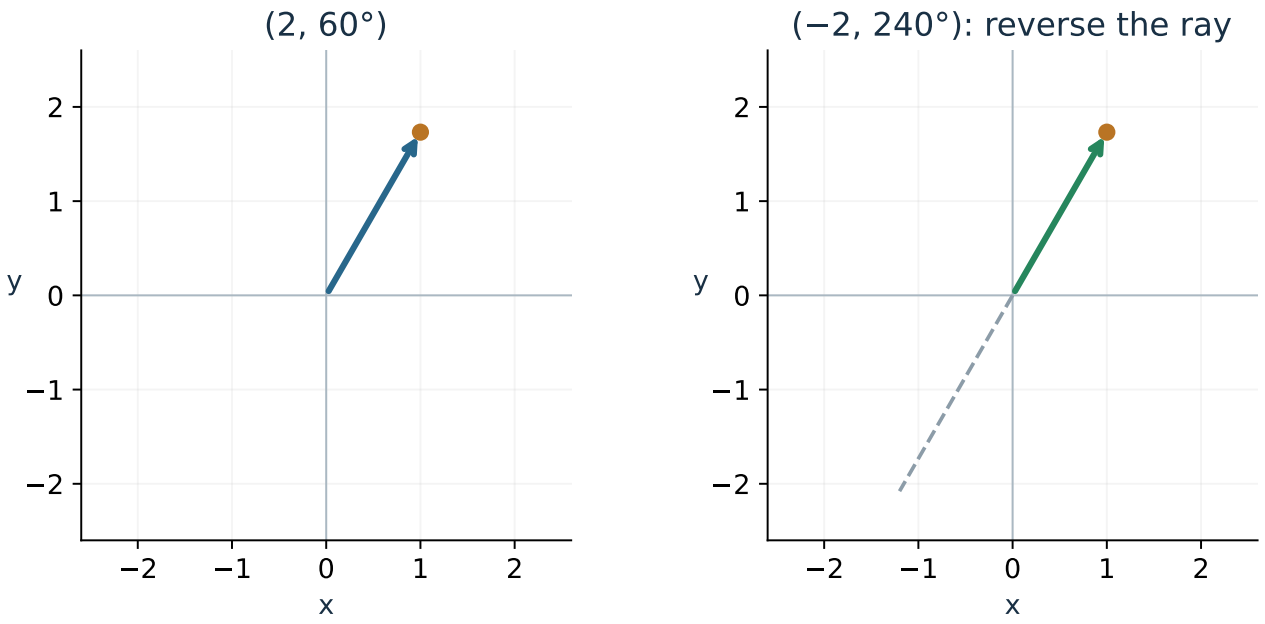


Figure 24: A positive radius on one ray and a negative radius on the opposite ray reach the same location.

For $z \neq 0$, an **argument** is an angle locating its direction. Polar form writes

$$z = r(\cos\theta + i\sin\theta), \qquad r = |z|.$$

Arguments differ by whole turns. The number zero has modulus zero and no defined argument.

Worked example: rectangular to polar form. For $z = -\sqrt{3} + i$, the modulus is $\sqrt{3+1} = 2$. The point lies in Quadrant II with reference angle $\pi/6$, so one argument is $5\pi/6$:

$$z = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right).$$

Multiplying out the two components returns the original number.

7.4 Addition combines components; multiplication combines transformations

Complex addition follows vector addition:

$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

For multiplication, expand and use $i^2 = -1$:

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

For example, $(1 + i)(1 - i) = 1 - i^2 = 2$, while $(1 + i) + (1 - i) = 2$ as well. These particular results agree, but the two operations follow different rules.

To see the geometric rule for multiplication, write

$$z = r(\cos\alpha + i\sin\alpha), \qquad w = s(\cos\beta + i\sin\beta).$$

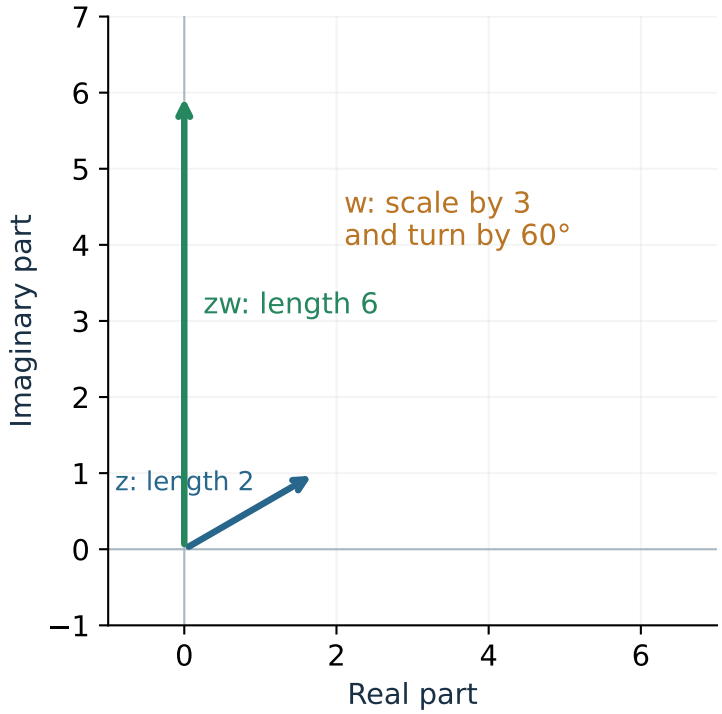


Figure 25: Multiplication by a complex factor rotates the original vector and scales its length.

Expanding gives real part $rs(\cos\alpha\cos\beta - \sin\alpha\sin\beta)$ and imaginary part $rs(\sin\alpha\cos\beta + \cos\alpha\sin\beta)$. The angle-addition identities turn these into

$$zw = rs\big(\cos(\alpha+\beta) + i\sin(\alpha+\beta)\big).$$

Multiply the moduli and add the arguments.

Worked example: rotate and stretch. Suppose z has modulus 2 and argument 30°, while w has modulus 3 and argument 60°. Then zw has modulus 6 and argument 90°, so $zw = 6i$.

Multiplication by i gives a quarter-turn counterclockwise: $i(a+bi) = -b + ai$, so (a, b) becomes $(-b, a)$. Its length stays fixed. Multiplication by -1 gives a half-turn. Multiplication by a positive real number changes length while keeping direction.

Use the Polar and Complex Explorer to predict a product's length and direction before displaying its rectangular coordinates.

7.5 Powers repeat the same rotation and scaling

Raising a number to a positive integer power repeats its multiplication rule:

$$\big[r(\cos\theta + i\sin\theta)\big]^n = r^n(\cos n\theta + i\sin n\theta).$$

This is **De Moivre's theorem**. It also holds for zero and negative integer powers when the original number is nonzero.

For example, $(\cos(\pi/6) + i\sin(\pi/6))^3$ has modulus 1 and angle $\pi/2$, so it equals i. The exponent multiplies the original angle by 3.

To find roots, reverse those two changes. Suppose

$$z^n = R(\cos\phi + i\sin\phi), \qquad R > 0.$$

The root modulus is $R^{1/n}$. For the angles, include all complete turns before dividing:

$$\theta_k = \frac{\phi + 2\pi k}{n}, \qquad k = 0, 1, \ldots, n-1.$$

The roots are equally spaced around a circle.

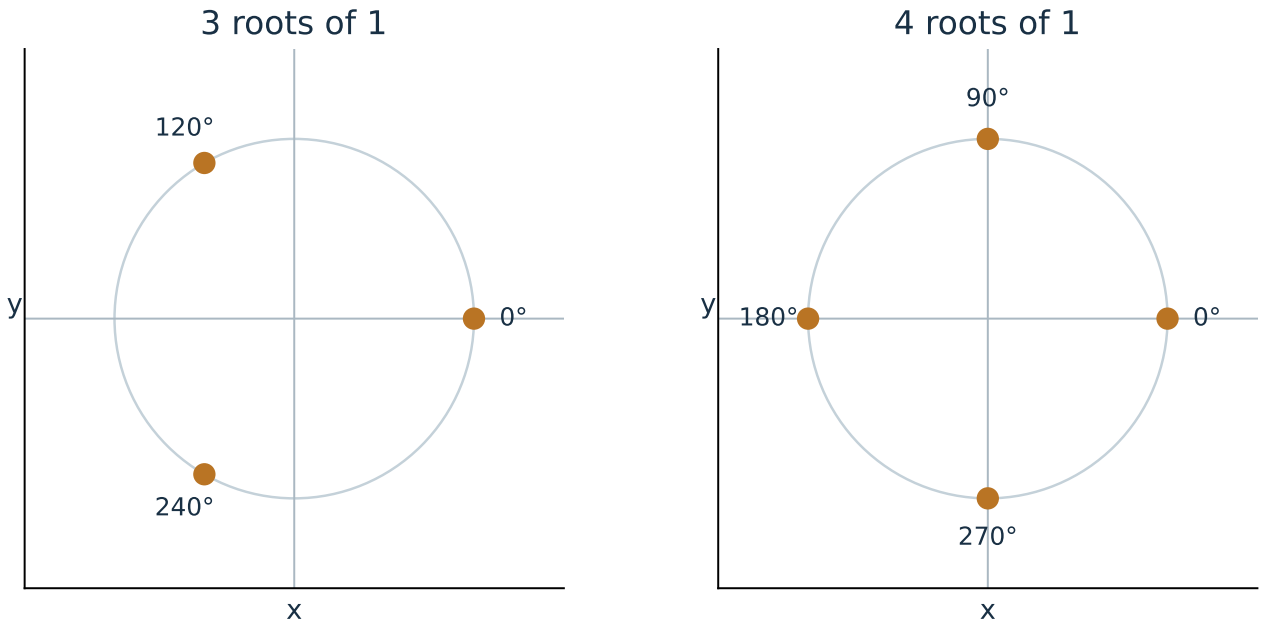


Figure 26: The cube roots of one are separated by 120 degrees; the fourth roots are separated by 90 degrees.

Worked example: three cube roots of 1. Use $R = 1, \phi = 0, n = 3$. The angles are $0, 2\pi/3, 4\pi/3$, giving

$$1, \qquad -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \qquad -\frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

Cubing any of these multiplies its angle by 3, producing a whole number of turns.

For the four fourth roots of 1, the angles are $0, \pi/2, \pi, 3\pi/2$, giving $1, i, -1, -i$.

Try it. Find all square roots of $4i$. Begin with modulus 4 and argument $\pi/2$.

7.6 A polar equation draws points by changing the radius

In a polar equation such as $r = 2\cos\theta$, each angle supplies a radius. Compute a few pairs, convert them to points, and look for the curve. When the formula supplies a negative radius, reverse the ray as described earlier.

Algebra identifies this particular curve. Multiply by r, then use $r^2 = x^2 + y^2$ and $r\cos\theta = x$:

$$r = 2\cos\theta \implies x^2 + y^2 = 2x \implies (x-1)^2 + y^2 = 1.$$

It is a circle with center $(1, 0)$ and radius 1. Multiplying by r deserves a check at $r = 0$; here the origin belongs to the original curve, since $\cos\theta = 0$ for suitable angles.

The curve is traced once as θ runs from 0 to π. From $\pi/2$ onward, the radius is negative, but the resulting points continue around the lower half of the same circle.

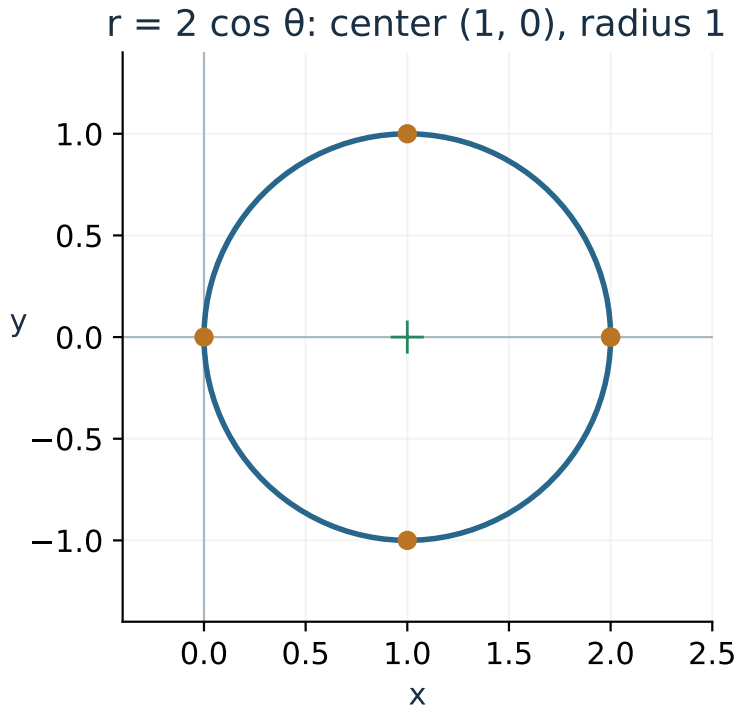


Figure 27: Sample polar coordinates for r equal to twice cosine trace a circle whose center is displaced from the origin.

7.7 Review exercises

1. Convert $(5, 2\pi/3)$ to Cartesian coordinates.
2. Give two distinct polar representations of $(\sqrt{2}, \sqrt{2})$, including one with negative radius.
3. Write $-1 - \sqrt{3}i$ in polar form with argument in $[0, 2\pi)$.
4. Calculate $(2 + i)(1 - 3i)$ in rectangular form.
5. Describe multiplication by $-i$ geometrically.
6. Evaluate $(\cos(\pi/4) + i\sin(\pi/4))^4$.
7. Find and plot all three cube roots of 1.
8. Convert $r = 4\sin\theta$ to Cartesian form and identify the curve.
9. Find all square roots of -9.
10. Explain what happens to modulus and argument when a nonzero complex number is squared.

7.8 Check your understanding

The first Try it gives $(3, -3\sqrt{3})$, with positive angle $5\pi/3$. The square roots of $4i$ have modulus 2 and arguments $\pi/4, 5\pi/4$; in rectangular form they are $\sqrt{2} + \sqrt{2}i$ and $-\sqrt{2} - \sqrt{2}i$.

8 Parametric Equations and Conic Sections

A curve shows where a moving object can go. To describe its motion, we also need to know where it is at each time. Parametric equations supply that information by letting both coordinates depend on one parameter.

We will use moving points to understand curves, then examine the distance rules defining circles, ellipses, parabolas, and hyperbolas. These are different ways to describe the same geometric objects.

8.1 Let two coordinates depend on one input

A parametric curve has the form

$$x = f(t), \qquad y = g(t).$$

Choose a value of t, calculate both coordinates, and plot their ordered pair. If t represents time, that pair is a position at that instant.

For $x = t + 2,\ y = 3t - 1$, the values $t = -1, 0, 1$ give $(1, -4), (2, -1), (3, 2)$. The point moves right and up as t increases. Solving $t = x - 2$ and substituting gives $y = 3x - 7$, so the path is a line.

Eliminating the parameter means finding an equation involving only x and y. Keep the original interval for t in mind. For $0 \le t \le 1$, the preceding equations trace only the segment from $(2, -1)$ to $(3, 2)$.

8.2 The same curve can carry different motions

For $x = \cos t,\ y = \sin t$, the point moves counterclockwise around the unit circle. At $t = 0$ it is $(1, 0)$; at $t = \pi/2$ it reaches $(0, 1)$.

Replace t by $2t$. The point still lies on the unit circle, because $\cos^2(2t) + \sin^2(2t) = 1$. It now completes a revolution in π units of time.

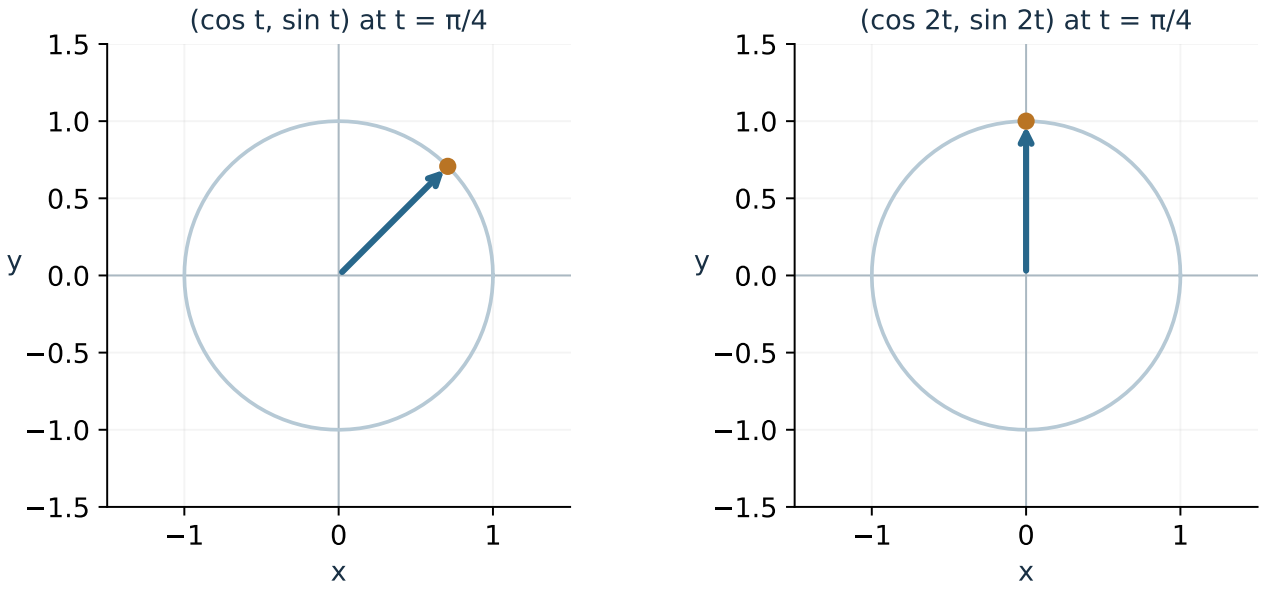


Figure 28: The two motions occupy the same circle but reach different positions at the same time.

On $0 \le t \le 2\pi$, $(\cos t, \sin t)$ travels around once, while $(\cos 2t, \sin 2t)$ travels twice. Replacing t by $-t$ reverses the direction. The Cartesian equation $x^2 + y^2 = 1$ describes all three paths, but contains no timing information.

Worked example: eliminate a trig parameter. Suppose

$$x = 4\cos t, \qquad y = 3\sin t.$$

Divide by the two scale factors, square, and add:

$$\left(\frac{x}{4}\right)^2 + \left(\frac{y}{3}\right)^2 = \cos^2 t + \sin^2 t = 1.$$

Thus the curve is $x^2/16 + y^2/9 = 1$. A full turn of t traces the whole ellipse; $0 \le t \le \pi$ traces its upper half.

Try it. Eliminate t from $x = 2t + 1,\ y = t - 3$, with $0 \le t \le 4$. Give both the line equation and the endpoints of the traced segment.

8.3 Follow a projectile with two equations

In an idealized projectile model, horizontal velocity stays constant while gravity changes the vertical motion. If the object begins at (x_0, y_0), with launch speed v_0 at angle θ, then

$$x(t) = x_0 + v_0\cos\theta\, t, \qquad y(t) = y_0 + v_0\sin\theta\, t - \frac{1}{2}gt^2.$$

Use $g = 9.8$ m/s^2 when distances are in meters and time is in seconds. The model ignores air resistance.

Worked example: a launch from the origin. Let $v_0 = 20$ m/s and $\theta = 30°$. Then

$$x(t) = 10\sqrt{3}\, t, \qquad y(t) = 10t - 4.9t^2.$$

At $t = 1$, the position is $(10\sqrt{3}, 5.1)$ meters. To find the landing time on level ground, set $y(t) = 0$:

$$t(10 - 4.9t) = 0.$$

The launch is at $t = 0$; landing occurs at $t = 100/49 \approx 2.04$ s. The physical trajectory uses that time interval.

Eliminating $t = x/(10\sqrt{3})$ produces a quadratic equation in x, so the path is a parabola. The time equations additionally tell us which direction the object moves and when it reaches each point.

8.4 A circle fixes one distance

A circle is the set of points a fixed distance $r > 0$ from a center (h, k). The distance formula gives

$$(x - h)^2 + (y - k)^2 = r^2.$$

For $(x - 2)^2 + (y + 1)^2 = 16$, the center is $(2, -1)$ and the radius is 4. A parameterization is $x = 2 + 4\cos t,\ y = -1 + 4\sin t$.

The signs inside the squares tell you which point makes both squared terms zero. This is a reliable way to read the center.

8.5 An ellipse fixes a sum of distances

An ellipse consists of points whose distances to two fixed **foci** add to a constant. Imagine a loop of string around two pins: holding the string taut with a pencil traces an ellipse.

For an ellipse centered at (h, k) with horizontal major axis,

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1, \qquad a > b > 0.$$

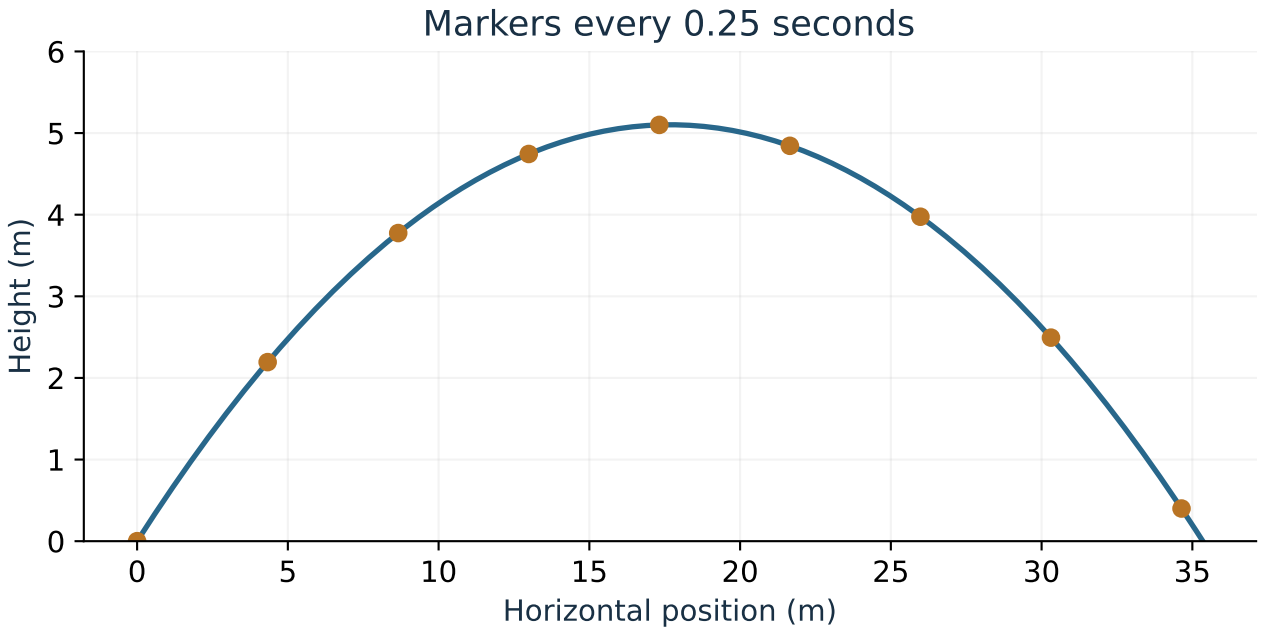


Figure 29: Equal time steps produce equal horizontal changes, while vertical changes vary under gravity.

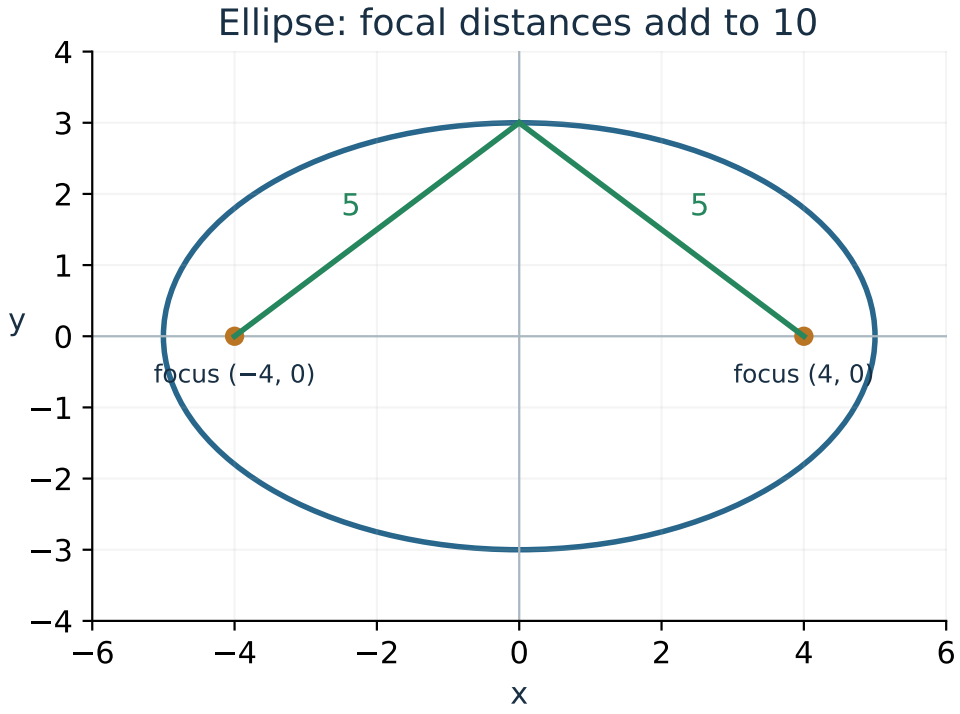


Figure 30: The two focal distances add to 2a at every point of the ellipse. At the top, the distance formula gives a squared equal to b squared plus c squared.

Its horizontal semiaxis has length a, and its vertical semiaxis length b. The foci are $(h-c,k),(h+c,k)$, with

$$c^2 = a^2 - b^2.$$

At a horizontal vertex, the two focal distances add to $2a$. At the top of the ellipse, symmetry makes each distance a; the right triangle there gives $a^2 = b^2 + c^2$.

Worked example: identify the ellipse. For $x^2/25 + y^2/9 = 1$, the semiaxes are 5 and 3. Thus $c = \sqrt{25-9} = 4$, and the foci are $(\pm 4, 0)$. Every point on the curve has total focal distance 10.

If the larger denominator is under the y-term, the major axis is vertical. Keep a for the larger semiaxis; then the foci move vertically from the center. Equal denominators give a circle, with coincident foci.

8.6 A parabola balances distance to a point and a line

A parabola consists of points equally far from a focus and a line called the **directrix**. For

$$(x-h)^2 = 4p(y-k), \qquad p \neq 0,$$

the vertex is (h,k), focus $(h,k+p)$, and directrix $y = k-p$. The sign of p determines whether the curve opens upward or downward.

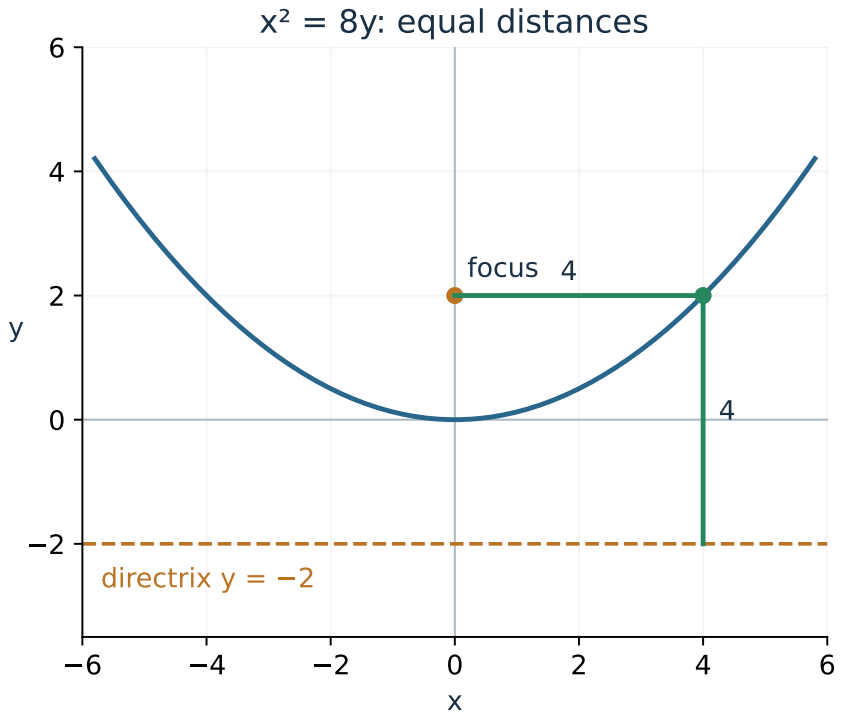


Figure 31: A point on the parabola is equally far from the focus and the directrix.

To derive the equation in the centered case, equate the distances from (x,y) to $(0,p)$ and to $y=-p$:

$$\sqrt{x^2 + (y-p)^2} = |y+p|.$$

Squaring gives $x^2 + (y-p)^2 = (y+p)^2$, which simplifies to $x^2 = 4py$.

For $x^2 = 8y$, $4p = 8$, so $p = 2$. The focus is $(0,2)$, and the directrix is $y = -2$. For $(x-1)^2 = 12(y+2)$, first locate the vertex $(1,-2)$; then $p = 3$, giving focus $(1,1)$ and directrix $y = -5$.

8.7 A hyperbola fixes a difference of distances

A hyperbola consists of points whose absolute difference of distances to two foci is constant. Its two branches lie near the two foci. For the horizontal form

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1,$$

the vertices are $(h \pm a, k)$, the foci are $(h \pm c, k)$, and

$$c^2 = a^2 + b^2, \qquad y - k = \pm\frac{b}{a}(x-h)$$

are the two asymptotes.

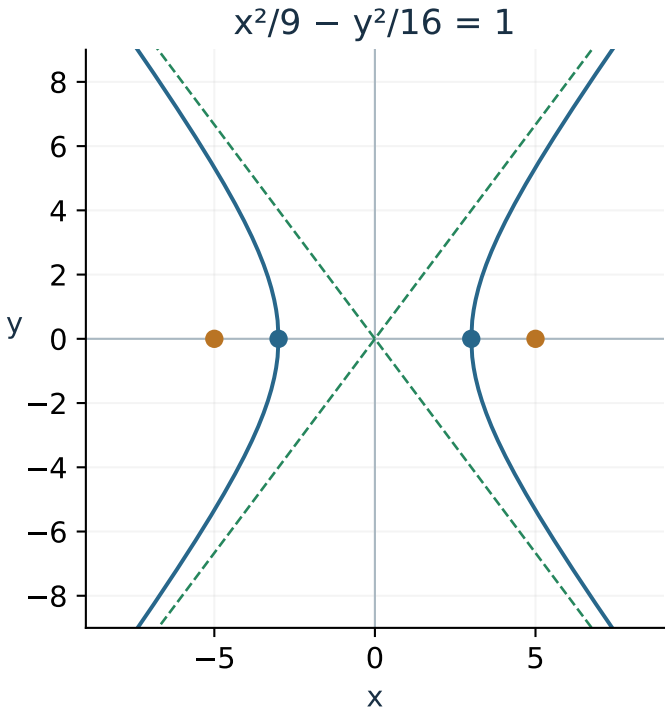


Figure 32: The branches of a hyperbola approach two asymptotes. Its foci lie farther from the center than its vertices.

For $x^2/9 - y^2/16 = 1$, we have $a = 3, b = 4, c = 5$. The vertices are $(\pm 3, 0)$, the foci $(\pm 5, 0)$, and the asymptotes $y = \pm 4x/3$. Far from the center, the constant 1 becomes small relative to the squared terms, explaining why the graph approaches those lines.

The sign difference is important. Ellipses use a sum of positive squared terms and remain bounded. Hyperbolas use a difference and extend along two branches.

8.8 Connect the descriptions

Use the Curves and Conics Explorer to move a point along a selected curve. Its slider records the fraction of the displayed path traversed. For circles and ellipses, one complete traversal corresponds to an angle change of 2π.

A parametric description helps with motion and plotting. A Cartesian equation helps identify centers, axes, and intercepts. A focal definition explains a distance relationship. Translate among these descriptions while keeping the parameter interval and orientation visible.

8.9 Review exercises

1. Eliminate t from $x = t + 2,\ y = 3t - 1$, and describe the motion.
2. Find the Cartesian equation for $x = 4\cos t,\ y = 3\sin t$.
3. Compare $(\cos t, \sin t)$ and $(\cos 3t, \sin 3t)$ on $[0, 2\pi]$.
4. Write the projectile equations for a launch from the origin at 20 m/s and 30°.
5. Identify center and radius of $(x-2)^2 + (y+1)^2 = 16$.
6. Find the foci of $x^2/36 + y^2/20 = 1$.
7. Find the focus and directrix of $(x-1)^2 = 12(y+2)$.
8. Find the vertices and asymptotes of $x^2/4 - y^2/9 = 1$.
9. Parameterize the circle of radius 3 centered at $(-2, 1)$.
10. Give a parameter interval tracing only the lower half of $x = 4\cos t,\ y = 3\sin t$.

8.10 Check your understanding

The Try it gives $y = (x-7)/2$, with endpoints $(1, -3)$ and $(9, 1)$. The line equation describes a whole line; the parameter interval selects this particular segment and the direction from its first endpoint to its second.

9 Sequences and Series

Suppose a theater adds two seats to each successive row. We can ask how many seats are in row 15, or how many seats are in the first 15 rows altogether. Those questions use the same pattern but ask for different quantities.

A sequence records individual values. A series adds them. We will build formulas from repeated addition and multiplication, keeping the meaning of each index and total in view.

9.1 Separate a term from a running total

A sequence assigns a number a_n to each integer index in a specified list. Usually we begin at $n = 1$, though a time model may begin at $n = 0$. The first terms of $a_n = 3 + 2(n - 1)$ are $3, 5, 7, 9, \ldots$.

The tenth term is a_{10}. The sum of the first ten terms is S_{10}:

$$S_n = a_1 + a_2 + \cdots + a_n.$$

The subscript tells you how far along the list to go. It does not mean multiplication.

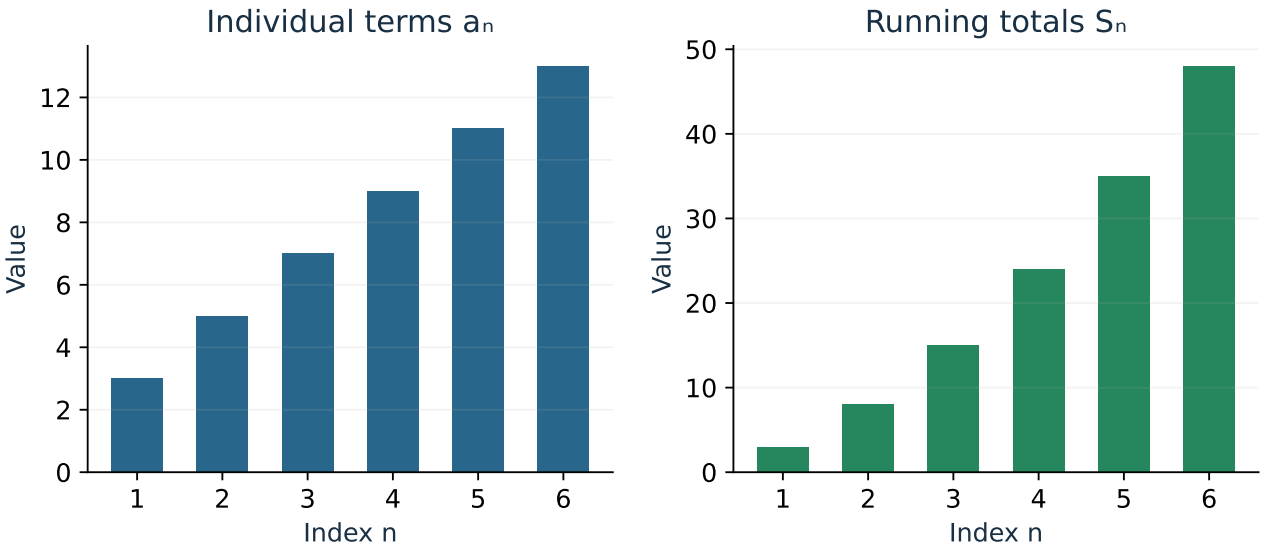


Figure 33: Individual terms of a sequence and the sums accumulated through those terms answer different questions.

For $3, 5, 7, 9, \ldots$, the first four partial sums are $3, 8, 15, 24$. The fourth term is 9, while the fourth partial sum is 24.

Try it. List the first five terms of $a_n = 7 - 3(n - 1)$, then calculate their sum. Some terms are negative; include their signs.

9.2 Repeated addition produces an arithmetic sequence

An **arithmetic sequence** adds the same common difference d at each step:

$$a_{n+1} = a_n + d.$$

Starting from a_1, reaching term n requires $n - 1$ additions. Therefore

$$a_n = a_1 + (n - 1)d.$$

The recursive rule tells you how to take a step; the explicit formula lets you jump directly to any term.

Worked example: theater rows. The first row has 12 seats, and each next row has 2 more. Row 15 has

$$a_{15} = 12 + (15 - 1)(2) = 40$$

seats. There are 14 increases between the first and fifteenth rows.

To add the rows, write the sequence forward and backward:

$$S_n = a_1 + a_2 + \cdots + a_n,$$
$$S_n = a_n + a_{n-1} + \cdots + a_1.$$

Each column adds to $a_1 + a_n$, and there are n columns. Thus

$$S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}(2a_1 + (n - 1)d).$$

For the first 15 rows, the average of the endpoint counts is $(12 + 40)/2 = 26$. Multiplying by 15 gives 390 seats. The formula also works for an odd number of terms: the middle term equals that same average.

9.3 Repeated multiplication produces a geometric sequence

A **geometric sequence** multiplies by the same ratio q at each step:

$$a_{n+1} = qa_n, \qquad a_n = a_1 q^{n-1}.$$

For $a_1 = 80, q = 1/2$, the terms are $80, 40, 20, 10, 5, \ldots$. With $0 < q < 1$, positive terms decrease toward zero. With $q > 1$, positive terms grow. Negative ratios alternate signs.

A percentage change specifies a multiplication factor. An 8% increase multiplies by 1.08; a 12% decrease multiplies by 0.88.

Worked example: define the initial index. A population begins at 500 and grows 8% per generation. If $P_0 = 500$ records the initial population, then

$$P_n = 500(1.08)^n.$$

If the initial observation is labeled $a_1 = 500$, the formula is $a_n = 500(1.08)^{n-1}$. The physical model is the same; the labels differ by one.

9.4 Add a finite geometric pattern

Write a partial sum and its multiple by q:

$$S_n = a_1 + a_1 q + \cdots + a_1 q^{n-1},$$
$$qS_n = \qquad a_1 q + \cdots + a_1 q^{n-1} + a_1 q^n.$$

Subtracting cancels every middle term:

$$(1 - q)S_n = a_1 - a_1 q^n, \qquad S_n = a_1 \frac{1 - q^n}{1 - q} \quad (q \neq 1).$$

For $q = 1$, all terms equal a_1, so the total is na_1.

Worked example: a finite total. For $5, 3, 1.8, \ldots$, the ratio is 0.6. The sum through term 5 is

$$S_5 = 5\frac{1 - 0.6^5}{1 - 0.6} = 11.528.$$

This adds five terms. The fifth term alone is $a_5 = 5(0.6)^4 = 0.648$.

The finite-sum formula works for growing, decreasing, or alternating geometric patterns, provided $q \neq 1$. It does not require convergence.

9.5 Read sigma notation as an instruction to add

The notation

$$\sum_{k=1}^{4}(2k+1)$$

says: substitute $k = 1, 2, 3, 4$, then add the four results. It becomes $3 + 5 + 7 + 9 = 24$.

The letter k is an index used inside the instruction. The bottom and top bounds determine which values to use. For $\sum_{k=0}^{5} 2(3/2)^k$, there are six terms, since both endpoints count.

Before applying a formula, write the first few terms and the last term. This checks the first value, ratio, and number of terms.

9.6 Can infinitely many additions have a finite total?

For the geometric pattern with $|q| < 1$, the power q^n approaches zero. The finite-sum formula then approaches

$$\sum_{n=0}^{\infty} aq^n = \frac{a}{1-q}.$$

An infinite sum means the limit of its partial sums. We calculate finite totals and ask what value they approach.

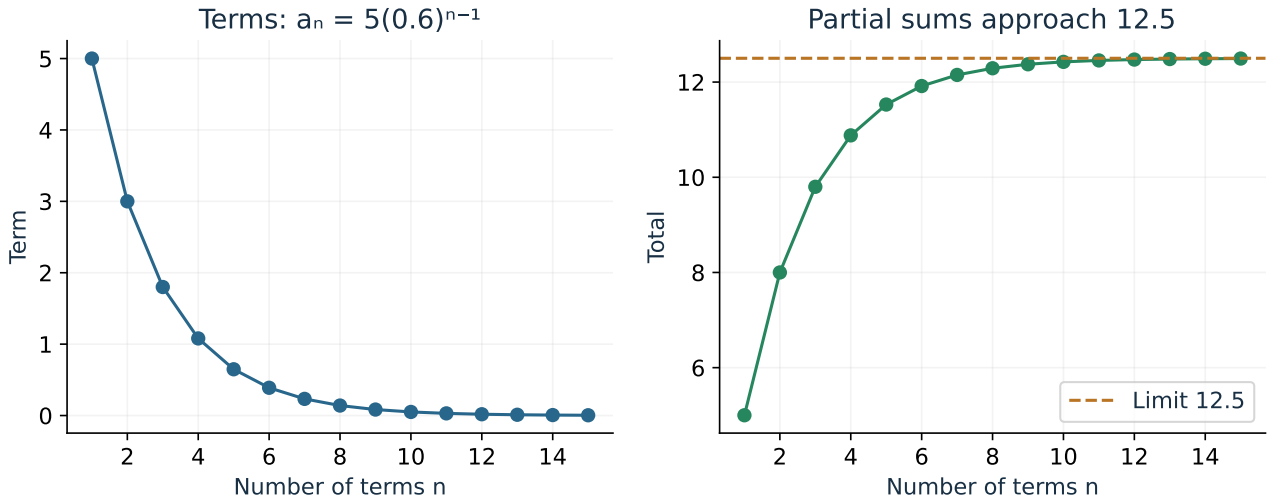


Figure 34: Geometric terms approach zero while their running totals approach a finite value. Ratios with magnitude at least one behave differently for nonzero first terms.

For $5 + 3 + 1.8 + \cdots$, the limiting total is $5/(1 - 0.6) = 12.5$. After n terms, the remaining amount is

$$\frac{5(0.6)^n}{1-0.6}.$$

This also measures how close a partial sum is to the limit.

Worked example: idealized bouncing. Drop a ball from 5 m, with each rebound reaching 60% of the previous height. The first drop contributes 5 m. Each later rebound height contributes twice, once upward and once downward:

$$5 + 2(5)(0.6 + 0.6^2 + \cdots) = 5 + 10\frac{0.6}{1-0.6} = 20 \text{ m}.$$

The model describes infinitely many idealized rebounds with a finite total travel distance.

For a geometric series with nonzero first term, $|q| \geq 1$ prevents convergence. With $q = 2$, terms grow; with $q = -1$, partial sums oscillate. Also, terms approaching zero alone do not guarantee convergence for every kind of series; the conclusion above follows from the geometric formula.

Try it. Find the sum of $4 - 2 + 1 - \frac{1}{2} + \cdots$. Identify its ratio and explain why the alternating signs are allowed.

9.7 A feedback rule can hide a geometric pattern

Consider

$$a_{n+1} = \frac{a_n + 6}{2}, \qquad a_1 = 0.$$

Each step averages the current value with 6. The first terms are $0, 3, 4.5, 5.25, 5.625, \ldots$. The changes are not constant, and neither are the ratios of consecutive terms.

Find the equilibrium by asking which value would remain fixed:

$$L = \frac{L + 6}{2} \quad \Longrightarrow \quad L = 6.$$

Now examine the difference $e_n = a_n - 6$. Subtracting 6 from the recurrence gives

$$e_{n+1} = \frac{e_n}{2}.$$

The **difference from equilibrium** is geometric, even though the original sequence is not. Since $e_1 = -6$,

$$a_n = 6 - 6\left(\frac{1}{2}\right)^{n-1}.$$

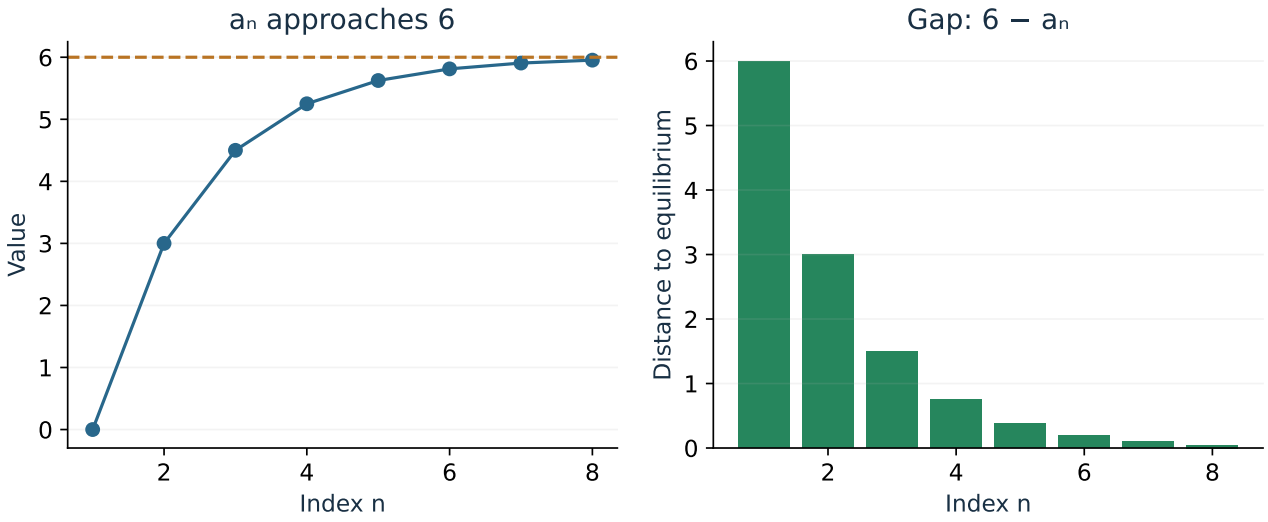


Figure 35: A feedback sequence approaches its equilibrium while the remaining gap halves at every step.

Use the Sequence Explorer to compare terms and partial sums of arithmetic and geometric sequences. For the feedback example, compute the terms by hand and use the formula for the remaining gap to explain the approach.

9.8 Review exercises

1. List five terms of $a_n = 7 - 3(n-1)$, and find a_{20}.
2. For $a_1 = 4, d = 5$, calculate S_{30}.
3. For $a_1 = 81, q = 1/3$, find a_6.
4. Compute $\sum_{k=0}^{5} 2(3/2)^k$ exactly.
5. Explain the difference between $\sum_{k=1}^{n} k$ and its final term n.
6. Find the sum of $5 + 5/2 + 5/4 + \cdots$.
7. A machine initially worth \$12,000 loses 12% of its value each year. Write its value after n years.
8. For $a_{n+1} = (a_n + 10)/2,\ a_1 = 2$, find the equilibrium and an explicit formula.
9. A theater has 20 rows, with 14 seats in the first and 3 more in each next row. Find the total.
10. Explain why $1 - 1 + 1 - 1 + \cdots$ has no limiting partial sum.

9.9 Check your understanding

The first Try it gives $7, 4, 1, -2, -5$, with sum 5. The alternating geometric series has ratio $-1/2$ and sum $4/(1 + 1/2) = 8/3$. Its partial sums approach the limit from alternating sides.

10 Synthesis: Motion and Patterns

Imagine a light attached to a rotating wheel. We can record its angle, plot its coordinates, describe its position as a complex number, or take a photograph every few seconds. Each description records a different aspect of one motion.

This chapter brings those descriptions together. We will first follow a continuous rotation, then sample it, and finally combine repeated rotation with shrinking or growing distances. The goal is to choose and connect the tools developed throughout the course.

10.1 Begin with a physical description

A point moves counterclockwise on a circle of radius 5 m, completing one turn every 20 s. It begins at the rightmost point.

The angular speed is one full turn divided by the time per turn:

$$\omega = \frac{2\pi}{20} = \frac{\pi}{10} \text{ rad/s}.$$

At time t, the angle is $\theta(t) = \pi t/10$. If the point had begun at another angle θ_0, we would add that initial angle.

The unit-circle formulas give the position:

$$x(t) = 5\cos\left(\frac{\pi t}{10}\right), \qquad y(t) = 5\sin\left(\frac{\pi t}{10}\right).$$

Polar coordinates record $(r, \theta) = (5, \pi t/10)$, while complex form records

$$z(t) = 5\left(\cos\frac{\pi t}{10} + i\sin\frac{\pi t}{10}\right).$$

All three describe a position 5 m from the origin at the same angle.

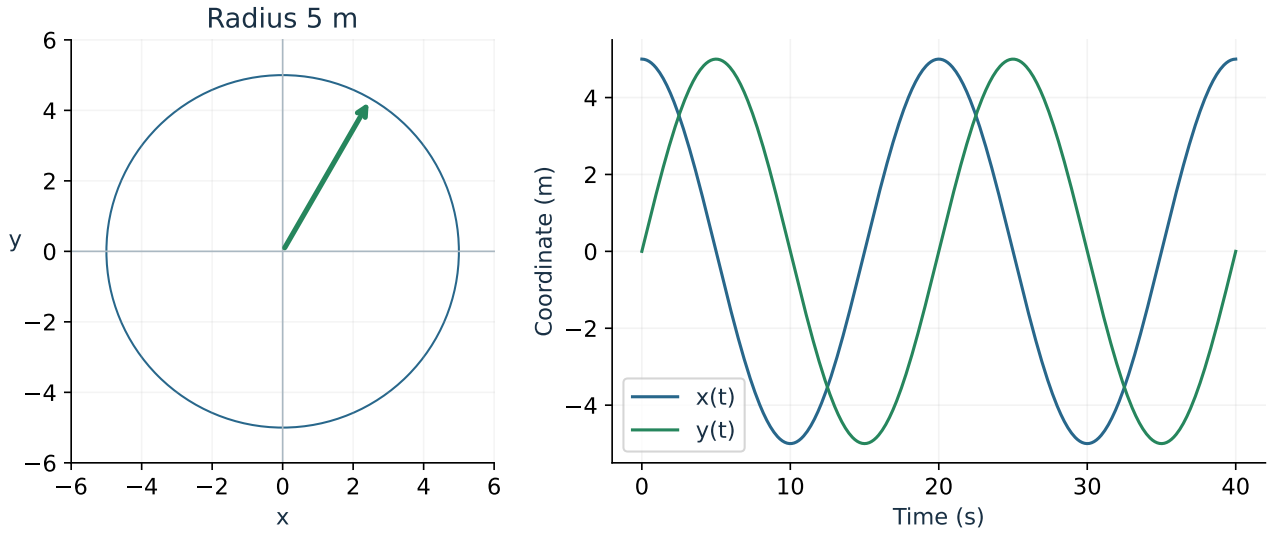


Figure 36: A point on a circle produces horizontal and vertical coordinate waves with the same period.

The coordinate waves have amplitude 5 and period 20 s. The circle itself has equation $x^2 + y^2 = 25$. That equation records the path; the time formulas additionally record orientation and timing.

10.2 Compare position, displacement, and distance traveled

After 5 seconds, the point moves from $(5, 0)$ to $(0, 5)$. Its displacement is

$$\Delta \mathbf{r} = \langle 0 - 5, 5 - 0 \rangle = \langle -5, 5 \rangle,$$

with magnitude $5\sqrt{2}$ m.

The distance traveled is a quarter of the circumference:

$$s = 5\left(\frac{\pi}{2}\right) = \frac{5\pi}{2} \text{ m}.$$

The curved distance exceeds the straight displacement magnitude. The average velocity over this interval is the displacement divided by elapsed time, $\langle -1, 1 \rangle$ m/s. The average speed is distance divided by time, $\pi/2$ m/s.

For constant angular speed, speed along the circle is $R|\omega|$. The instantaneous direction is tangent to the circle. We can recognize that geometric direction without differentiating the coordinate formulas.

Try it. For this same motion, compare displacement and distance traveled over a complete 20-second turn.

10.3 Sampling turns continuous motion into a sequence

Suppose a camera records the position every 5 seconds. The angle advances by

$$\delta = \omega \Delta t = \frac{\pi}{10}(5) = \frac{\pi}{2}$$

between photographs. Beginning with index $n = 0$, the first five positions are

$$z_0 = 5, \quad z_1 = 5i, \quad z_2 = -5, \quad z_3 = -5i, \quad z_4 = 5.$$

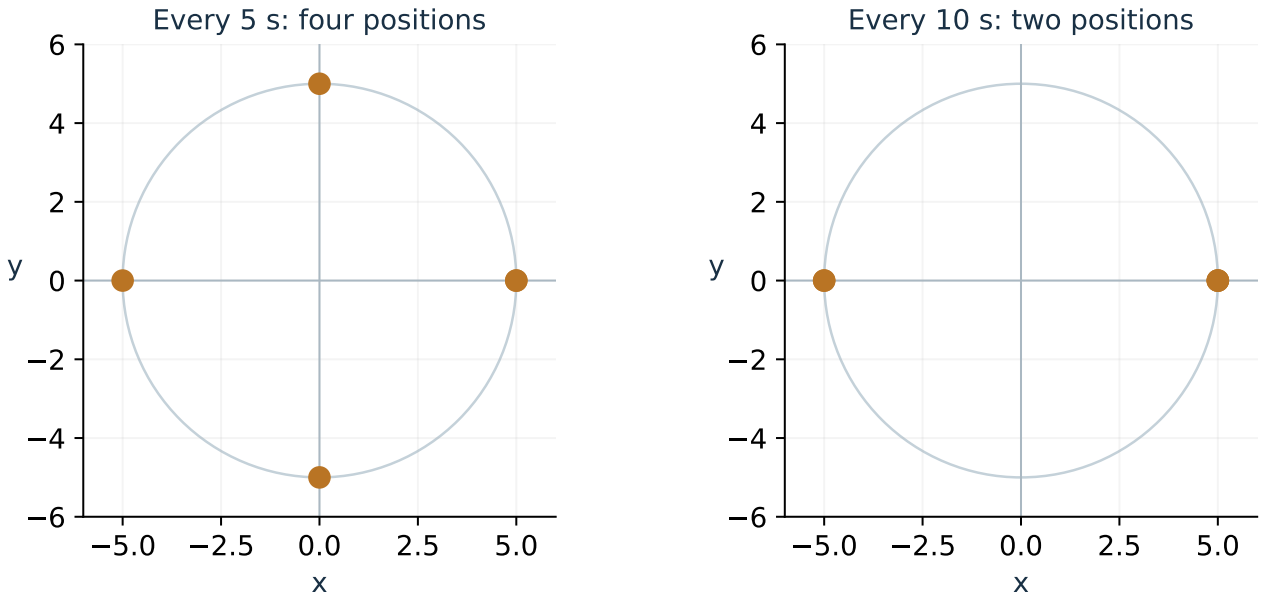


Figure 37: Sampling every quarter-turn yields four distinct positions; other sampling intervals can reveal different patterns of the same motion.

Each sample is the previous one multiplied by i. We have both a recurrence and an explicit formula:

$$z_{n+1} = iz_n, \qquad z_n = 5i^n.$$

De Moivre's theorem expresses the same result as

$$z_n = 5\big(\cos(n\pi/2) + i\sin(n\pi/2)\big).$$

More generally, a fixed angle step δ corresponds to

$$u = \cos\delta + i\sin\delta, \qquad z_{n+1} = uz_n.$$

Repeated multiplication gives $z_n = u^n z_0$.

The sampling interval matters. Photographing the original wheel every 20 seconds would always capture the same point. Those photographs alone would conceal the motion between them. Sampling every 10 seconds gives two alternating positions.

10.4 Decide whether sampled positions repeat

For a nonzero starting point and unit-length multiplier, a sample returns to its start when the accumulated angle is a whole number of turns:

$$n\delta = 2\pi k$$

for integers $n > 0, k$.

If $\delta/(2\pi) = p/q$ in lowest terms with $q > 0$, the first return is after q steps. For example, $\delta = 4\pi/5$ is $2/5$ of a turn, so five steps close the sampled orbit. The points are visited in a star order rather than around consecutive edges.

If $\delta/(2\pi)$ is irrational, no positive integer number of steps gives an exact return. A finite computer display can look almost closed, so distinguish a close approach from mathematical equality.

10.5 Add a scale change at each step

Now let the multiplier have modulus $\rho > 0$:

$$q = \rho(\cos\delta + i\sin\delta), \qquad z_{n+1} = qz_n.$$

After n steps,

$$z_n = q^n z_0, \qquad |z_n| = \rho^n |z_0|.$$

The angle advances by $n\delta$, measured modulo full turns. The distance from the origin follows a geometric sequence.

For $0 < \rho < 1$, radii shrink toward zero. For $\rho = 1$, they stay constant. For $\rho > 1$, they grow. These statements describe a nonzero starting point. The origin stays fixed.

Worked example: track three shrinking quarter-turns. Start at $z_0 = 4$ and multiply by $q = i/2$:

$$z_1 = 2i, \qquad z_2 = -1, \qquad z_3 = -\frac{1}{2}i.$$

The radius halves each step: $4, 2, 1, 1/2$. The angle adds $90°$: $0°, 90°, 180°, 270°$. The explicit formula is

$$z_n = 4\left(\frac{1}{2}\right)^n \big(\cos(n\pi/2) + i\sin(n\pi/2)\big).$$

The angular pattern repeats, while the positions do not: their nonzero magnitudes keep decreasing.

Try it. Begin with $z_0 = 2$ and use $q = \frac{1}{2}(\cos 60° + i\sin 60°)$. Find the modulus, direction, and rectangular value of z_3.

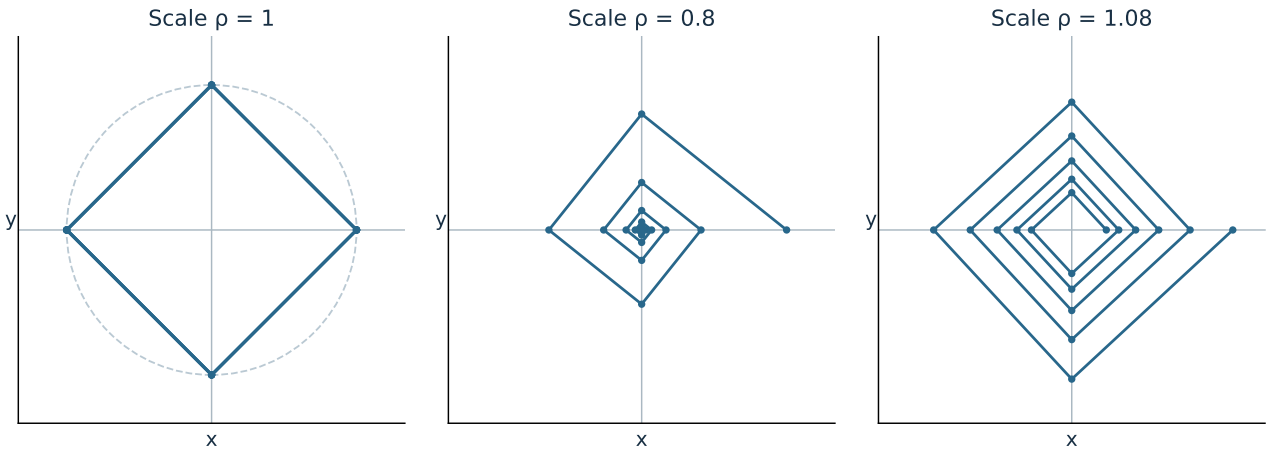


Figure 38: A repeated quarter-turn stays on a circle when the scale is one, spirals inward for a scale below one, and outward for a scale above one.

10.6 Adding step vectors builds a different sequence

The recurrence above multiplies the **position** at each step. Another model specifies the **displacement** during each step. If those steps are $\Delta z_n = aq^n$, starting from position z_0, then

$$z_N - z_0 = \sum_{n=0}^{N-1} aq^n = a\frac{1-q^N}{1-q} \qquad (q \neq 1).$$

The geometric-sum algebra also works for complex numbers, since multiplication and subtraction obey the same distributive rules.

For steps $1, i, i^2, i^3$, the total displacement is $1+i-1-i = 0$. From the origin, the running positions are $0, 1, 1+i, i, 0$. Those positions are the corners of a square.

For shrinking steps $a(i/2)^n$, the step lengths approach zero. Their total displacement approaches $a/(1-i/2)$. The final position depends on both that sum and the starting point. Always decide whether a formula describes locations or movements between locations.

10.7 Choose a representation that answers the question

A model begins with quantities and units. For rotating machinery, angle and period may be the natural measurements. For navigation, components help add movements. For repeated scaling and rotation, complex multiplication gives a compact rule. For a path with a particular timing, parametric equations supply both coordinates at each instant.

These choices also reveal the assumptions. A constant-speed circle needs a fixed radius and angular speed. An ellipse has a different relation between its horizontal and vertical coordinates. An idealized projectile uses constant horizontal velocity and downward acceleration. Select a model because its assumptions fit the question.

Use the Motion Synthesizer to vary the scale and angle independently. Fix one control while investigating the other. Then explain the plotted pattern through its radius sequence and angle sequence.

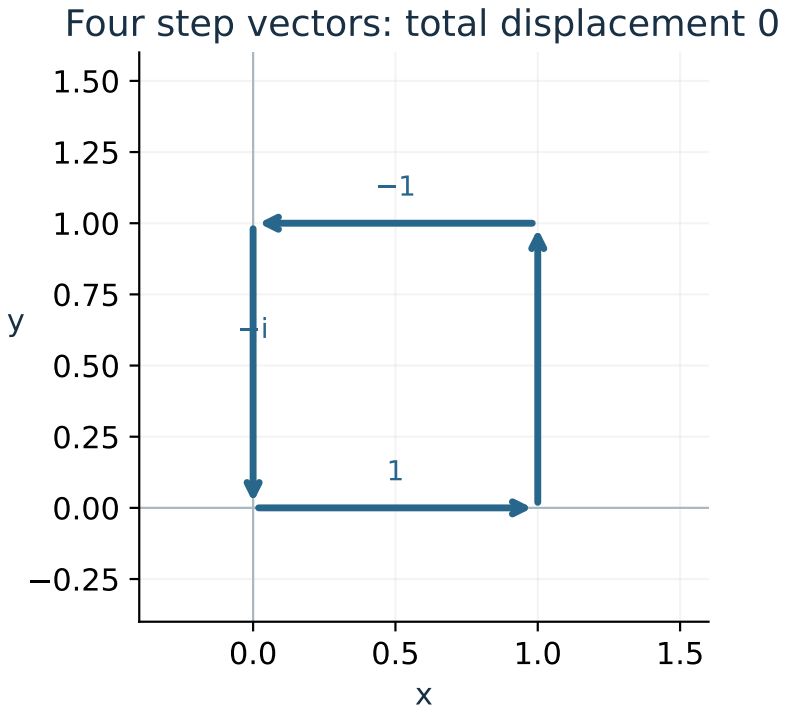


Figure 39: Adding four quarter-turn step vectors makes a closed polygon. The step vectors are different from the running positions.

10.8 Review exercises

1. Write Cartesian, polar, and complex formulas for radius 5 and period 20 s, starting on the positive horizontal axis.
2. Sample every 5 s. Give the first five complex positions and the repetition period in samples.
3. For $z_{n+1} = iz_n,\ z_0 = 1 + i$, find $z_1, \ldots, z_4$.
4. For $z_{n+1} = (i/2)z_n,\ z_0 = 4$, find z_4 and describe its magnitude and direction.
5. Eliminate t from $x = 3\cos t,\ y = 2\sin t$.
6. Explain why a circle can repeat while a shrinking spiral from a nonzero point does not.
7. Interpret $1 + i + i^2 + i^3$ as four step vectors.
8. Describe one process with at least three mathematical representations from the course.
9. A circle is sampled with angle step $4\pi/5$. How many steps give its first return?
10. Explain what photographs every full revolution would reveal and what information they would miss.

10.9 Check your understanding

Over a complete turn, displacement is zero and traveled distance is 10π m. In the shrinking $60°$ example, after three steps the modulus is $2(1/2)^3 = 1/4$, the direction is $180°$, and $z_3 = -1/4$.